
A DETERMINISTIC RECONSTRUCTION OF FINITE-DIMENSIONAL QUANTUM MECHANICS

A PREPRINT

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May 10, 2026

ABSTRACT

This paper reconstructs the operational content of finite-dimensional quantum mechanics from deterministic Hamiltonian dynamics defined on a compact Kähler manifold Σ . An ontic microstate evolves under a Liouville-preserving flow generated by a Hamiltonian vector field, while observable statistics arise only after applying a smooth many-to-one projection $\pi : \Sigma \rightarrow \mathbb{CP}^{n-1}$ to epistemic coordinates. The induced epistemic measure is taken to be absolutely continuous with respect to the Fubini–Study measure, which is selected as the canonical invariant density under the symmetry and regularity assumptions developed in companion work.

For the quantum-effective sector defined by Axiom A5, the projected dynamics on $\mathbb{CP}^{n-1}$ defines a holomorphic vector field. By the Ashtekar–Schilling correspondence, this yields the standard Schrödinger form of unitary evolution on the operational state space. Measurement interactions are modelled as Hamiltonian evolution on a joint system–apparatus surface, with outcomes represented by context-dependent measurable regions on $\mathbb{CP}^{n-1}$. Outcome probabilities are assigned by integrating the prepared epistemic density over these regions; in the finite-dimensional setting, the admissible Born-form weight assignment is fixed in companion work, while the mixed-state, POVM, and reduced-state representation layers are completed in TN4. In the projective examples studied here, the outcome regions admit explicit Fubini–Study Voronoi realisations.

Accordingly, the present paper should be read as the finite-dimensional reconstruction architecture for CSD: it specifies the ontic projection framework, the quantum-effective dynamical sector, and the context-to-region measurement map, while relying on companion results for the Born-weight theorem and the full operational dictionary.

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1 Introduction

1.1 Background

Standard finite-dimensional quantum mechanics is formulated on a Hilbert space $\mathcal{H} \cong \mathbb{C}^n$, with pure states represented by rays $[\psi] \in \mathbb{CP}^{n-1}$ and probabilities given by the Born rule

$$P(i) = |\psi_i|^2.$$

Although operationally successful, this axiomatic structure does not explain why the Hilbert space geometry, the Fubini–Study measure μ_{FS} , or the Born weights arise from underlying physical principles. Measurement is introduced as an external postulate, and no mechanism is provided for the emergence of tensor-product structure, decoherence, or contextuality. This motivates a reconstruction of quantum mechanics based on geometric and dynamical principles rather than algebraic axioms.

The conventional formalism treats the state vector ψ as fundamental, with its linear structure and probabilistic interpretation assumed rather than derived. The Born rule is postulated independently of any ontic structure. Measurement outcome geometry is not connected to physical interactions, and the composite-system tensor product has no geometric justification. These limitations motivate an ontic framework in which statistical structure emerges from dynamics on an underlying symplectic manifold (Σ, ω) .

Results from Paper A: Deterministic Frequency Generation: Paper A demonstrates that deterministic Hamiltonian evolution on a symplectic surface (Σ, ω) endowed with Liouville measure μ_L yields stable long-run frequencies determined by invariant volume. For any measurable partition $\{\Omega_i\}$ of Σ induced by a measurement interaction, the empirical frequencies satisfy

$$f_i \longrightarrow \frac{\mu_L(\Omega_i)}{\mu_L(\Omega_0)}$$

for almost all initial conditions, where Ω_0 is the invariant region explored by the flow. This result does not require stochastic assumptions or collapse postulates, and establishes invariant volume as the statistical foundation of CSD.

Results from Paper B: Uniqueness of the Epistemic Measure: Paper B proves that any epistemic measure on $\mathbb{CP}^{n-1}$ compatible with complex rescaling $\psi \mapsto \lambda\psi$ and covariance under $\text{SU}(n)$ must be proportional to the Fubini–Study measure μ_{FS} . Therefore the pushforward of the Liouville measure satisfies

$$\pi_*\mu_L = \rho d\mu_{\text{FS}},$$

where the density ρ is fixed by symmetry. This establishes $|\psi_i|^2$ as a consequence of the geometric symmetries of $\mathbb{CP}^{n-1}$ rather than an independent probabilistic postulate.

Results from Paper D: Ontic–Epistemic Structure and Measurement Geometry: Paper D introduces the ontic–epistemic split central to Constraint Surface Dynamics. A physical microstate is a point $x \in \Sigma$ evolving under the Liouville-preserving flow generated by X_H . Observers access only the epistemic state $[\psi] = \pi(x)$, where $\pi : \Sigma \rightarrow \mathbb{CP}^{n-1}$ is a smooth many-to-one projection. Measurement interactions define outcome regions

$$\{\Omega_i(M)\} \subset \mathbb{CP}^{n-1},$$

realised in the projective examples by Fubini–Study Voronoi partitions determined by the apparatus interaction Hamiltonian. The boundaries of these regions have μ_{FS} -measure zero, ensuring consistent statistics. Paper D further shows that contextuality arises from geometric incompatibility between partitions associated with different observables.

Combining these results yields:

1. Deterministic Hamiltonian flow on Σ produces empirical frequencies through volume typicality (Paper A).
2. The projection π defines epistemic states whose measure is fixed by symmetry (Paper B).
3. Measurement outcome regions $\{\Omega_i(M)\}$ arise from the apparatus Hamiltonian through Fubini–Study geometry (Paper D).
4. Probabilities are obtained by integrating the prepared epistemic density over the outcome region,

$$P(i \mid M, \rho_{ep}) = \int_{\Omega_i(M)} \rho_{ep} d\mu_{\text{FS}}.$$

For a sharply prepared pure state, the canonical idealisation used in this paper is the rank-one density operator

$$\hat{\rho}_{[\psi]} = |\psi\rangle\langle\psi|.$$

Probabilities are then computed as

$$P(i \mid \hat{\rho}_{[\psi]}, M) = \text{Tr}(\hat{\rho}_{[\psi]} \Pi_i(M)),$$

or equivalently, when an epistemic density representation is used,

$$P(i \mid M, \rho^{ep}) = \int_{\Omega_i(M)} \rho^{ep} d\mu_{\text{FS}}.$$

The approach developed in this paper should be understood as a reconstruction rather than a derivation. We begin with a compact Kähler manifold Σ , a projection map $\pi : \Sigma \rightarrow \mathbb{CP}^{n-1}$, and the Fubini–Study geometry of projective Hilbert space as structural assumptions. These axioms do not claim uniqueness for the physical world; rather, they identify the minimal geometric requirements sufficient to reproduce the operational structure of finite-dimensional quantum mechanics. The objective of this work is therefore to establish internal mathematical closure and empirical completeness within this reconstruction, while relativistic and field-theoretic extensions will be addressed separately in Paper Σ .

1.2 Aim of This Paper

The objective of this work is to show that the full operational formalism of finite-dimensional quantum mechanics arises from deterministic Hamiltonian dynamics on Σ together with the geometric structure of $\mathbb{CP}^{n-1}$.

The reconstruction developed here differs in important respects from standard interpretations of quantum mechanics. Unlike Bohmian mechanics, the ontic dynamics occur on a compact Kähler manifold Σ rather than in configuration space, and no guiding equation is introduced. Unlike Everettian interpretations, no branching ontology is assumed; instead, the epistemic state reflects coarse-grained information under the projection π . No stochastic collapse primitives are added, distinguishing the framework from GRW-type theories. While the formalism shares geometric structure with Geometric Quantum Mechanics, it supplements this with a lower-level deterministic dynamics whose projection

reproduces the quantum sector. The present work is therefore closest in spirit to ontological reconstruction programmes, though it differs by grounding measurement, decoherence, and Born-form probabilities in deterministic ontic dynamics together with context-dependent measurable partitions on CP^{n-1} , the canonical Fubini–Study epistemic measure, and the state-dependent probability rule developed in Papers A and B.

Throughout this paper we assume the existence of a smooth many-to-one projection

$$\pi : \Sigma \longrightarrow CP^{n-1},$$

with $[\psi] = \pi(x)$ defining the epistemic state of an observer.

For Hamiltonians on Σ whose fibre dependence is negligible within the quantum-effective sector (the quantum-effective Hamiltonians defined in Appendix I), the pushforward π_*X_H is a holomorphic vector field on CP^{n-1} . By the Ashtekar–Schilling correspondence, this induces the epistemic evolution

$$\dot{\psi} = -\frac{i}{\hbar} \hat{H}\psi,$$

thereby recovering Schrödinger dynamics from deterministic flow on Σ .

Measurement interactions are represented by Hamiltonian evolution on a joint ontic surface Σ_{SA} for system and apparatus. The induced epistemic outcome regions

$$\{\Omega_i(M)\} \subset \mathbb{CP}^{n-1}$$

are realised in the projective examples by Fubini–Study Voronoi partitions determined by the apparatus Hamiltonian. Probabilities are assigned by the canonical state-dependent rule

$$P(i | M, \rho^{ep}) = \int_{\Omega_i(M)} \rho^{ep} d\mu_{FS},$$

or, for sharply prepared pure states,

$$P(i | \hat{\rho}, M) = \text{Tr}(\hat{\rho} \Pi_i(M)).$$

This is consistent with the measure-theoretic result of Paper B and the typicality theorem of Paper A. No stochastic postulate or collapse rule is invoked.

Composite systems are represented by a joint ontic surface Σ_{AB} that is not the Cartesian product $\Sigma_A \times \Sigma_B$, in accordance with Axiom A6. The projection π_{AB} is compatible with marginal stability and locally reproduces the epistemic product structure on $CP^{n_A-1} \times CP^{n_B-1}$, providing the composite geometric framework within which entanglement and subsystem structure are analysed. The finite-dimensional reduced-state and mixed-state representation layers are completed in TN4.

The aim of this paper is therefore to establish the finite-dimensional reconstruction architecture: closed Schrödinger-form dynamics in the quantum-effective sector, context-dependent measurable outcome regions on CP^{n-1} , and the geometric representation of interference, decoherence, and composite structure. The Born-form probability layer is fixed by Paper B, while the full mixed-state, POVM, and subsystem-reduction dictionary is completed in TN4.

1.3 Problems Addressed

The reconstruction developed in this paper addresses three interconnected levels of structure: the ontic geometry on Σ , the epistemic dynamics on CP^{n-1} , and the operational predictions accessible to observers. Each level presents specific problems that must be resolved in order to derive the finite-dimensional quantum formalism from deterministic Hamiltonian dynamics.

Ontic Level: The first task is to specify the geometric conditions under which the ontic surface Σ can support the structures required for quantum phenomena. We assume that Σ is a finite-dimensional compact Kähler manifold equipped with symplectic form ω , complex structure J , and compatible metric g . The Hamiltonian vector field X_H generated by a real-valued Hamiltonian H must preserve the Liouville measure μ_L associated with ω . A central problem at the ontic level is to identify the conditions under which the pushforward of the ontic vector field,

$$\pi_*X_H,$$

is linear in the epistemic coordinates ψ and therefore capable of generating standard quantum dynamics. For Hamiltonians in the quantum-effective sector, the resulting projected flow is Hamiltonian up to controlled corrections of order $O(\epsilon)$, and in the ideal limit $\epsilon = 0$ it becomes holomorphic and admits a linear representation on $\mathbb{C}^n$.

Epistemic Level: The second task is to recover the full mathematical structure of finite-dimensional quantum mechanics from the projected dynamics on CP^{n-1} . This includes:

1. Deriving Schrödinger evolution as the projected Hamiltonian flow generated by $\pi_* X_H$.
2. Constructing the outcome regions $\{\Omega_i(M)\}$ as context-dependent measurable partitions, realised in the projective examples studied here by Fubini–Study Voronoi geometry.
3. Recovering the Born-form probability rule

$$P(i \mid M, \rho^{ep}) = \int_{\Omega_i(M)} \rho^{ep} d\mu_{FS},$$

with the sharp-state idealisation

$$P(i \mid \hat{\rho}, M) = \text{Tr}(\hat{\rho} \Pi_i(M)),$$

as justified by the volume-typicality result of Paper A together with the symmetry and operational constraints established in Paper B.

4. Implementing subsystem reduction through the marginal maps π_A arising from the joint projection π_{AB} , and obtaining mixed states through the induced marginal densities.
5. Explaining decoherence through geometric separation of environmental degrees of freedom on Σ , with pointer states identified by stability under Fubini–Study volume spreading.

These components demonstrate how the epistemic manifold CP^{n-1} acquires the full structure normally attributed to Hilbert space, including interference, contextuality, and entanglement.

Operational Level: The final task is to show that the deterministic reconstruction reproduces the operational predictions of finite-dimensional nonrelativistic quantum mechanics. For each measurement interaction M with outcome regions $\{\Omega_i(M)\}$, the theory must yield

$$P(i \mid M, \rho_{ep}) = \int_{\Omega_i(M)} \rho_{ep} d\mu_{FS},$$

consistent with the Born rule once ρ_{ep} encodes the preparation. Sequential and composite measurements must reduce to overlaps of Fubini–Study regions and marginal projections. The reconstruction must therefore demonstrate empirical equivalence with the operational content of standard finite-dimensional quantum mechanics without invoking collapse postulates or intrinsic stochasticity.

This section identifies the specific problems that the remainder of the paper resolves at the ontic, epistemic, and operational levels.

1.4 Scope and Limitations

The reconstruction developed in this paper applies only to finite-dimensional, nonrelativistic quantum systems. All results are formulated on the epistemic manifold CP^{n-1} and rely on the compactness and finite-dimensional Kähler geometry of the ontic surface Σ . The analysis does not apply to infinite-dimensional Hilbert spaces, continuous spectra, quantum fields, or continuous-mode systems, which require noncompact projective spaces and infinite-dimensional symplectic geometry.

The present work does not treat identical-particle sectors. Bosonic and fermionic symmetries require additional geometric structure that is developed separately in Paper Σ , where permutation symmetry and indistinguishability are incorporated into both the ontic and epistemic manifolds. The emergence of relativistic spacetime geometry is also not addressed. No relativistic structure is introduced in this paper, and the derivation of Lorentzian geometry, causal ordering, and relativistic dynamics is carried out in Paper Σ .

The reconstruction presented here assumes the following structural elements. First, there exists a compact Kähler manifold $(\Sigma, \omega_\Sigma, J_\Sigma, g_\Sigma)$ carrying a Liouville-preserving Hamiltonian flow X_H , together with a smooth many-to-one projection $\pi: \Sigma \rightarrow \text{CP}^{n-1}$ whose pushforward $\pi_* \mu_L$ is absolutely continuous with respect to the Fubini–Study measure μ_{FS} and compatible with $\text{SU}(n)$ symmetry. Second, at the epistemic level the analysis assumes only the results of Papers A and B: that long-run frequencies follow invariant volumes on Σ and that μ_{FS} is the unique symmetry-compatible measure on CP^{n-1} .

Given these assumptions, the remainder of the paper establishes the finite-dimensional reconstruction architecture used in CSD. More precisely, it shows how a closed Schrödinger-form epistemic dynamics arises in the quantum-effective

sector, how measurement contexts are represented by measurable outcome regions on $\mathbb{CP}^{n-1}$, and how contextual dependence is encoded geometrically. The present paper does not by itself prove every element of the finite-dimensional operational dictionary. The Born-form probability assignment is fixed in Paper B, and the mixed-state, POVM, and subsystem-reduction layers are completed in TN4. Likewise, the existence and regularity of Σ and π are assumed rather than derived, and the deeper physical origin of the quantum-effective sector selected by A5 is left for later work.

1.5 Structure of the Paper

The remainder of this paper is organised as follows. Section 2 introduces the ontic surface Σ , its Kähler structure, and the deterministic Hamiltonian flow governing microstate evolution. The conditions under which the projected dynamics becomes linear in the epistemic coordinates are identified, together with the role of quantum-effective Hamiltonians.

Section 3 develops the epistemic description on $\mathbb{CP}^{n-1}$. Schrödinger evolution is obtained as the projected Hamiltonian flow generated by $\pi_* X_H$. Measurement outcome regions are treated abstractly as context-dependent measurable partitions of $\mathbb{CP}^{n-1}$ satisfying the A7 regularity conditions. In the finite-dimensional projective examples analysed in this paper, these regions admit explicit Fubini–Study Voronoi realisations. This allows the geometric structure of measurement, conditionalisation, and contextual incompatibility to be studied concretely without identifying Voronoi geometry with the full abstract content of Axiom A7.

Section 4 examines composite systems. The joint ontic surface Σ_{AB} is not assumed to factorise, and the projection π_{AB} is analysed to determine how marginal stability and local epistemic product structure can be realised. The section develops the composite geometric framework and the representation of entanglement. The finite-dimensional proof that the induced subsystem representation reproduces the standard reduced-state formalism is supplied in TN4.

Section 5 presents the geometric measurement framework and the modelling of decoherence. Measurement interactions are treated as Hamiltonian flows on Σ_{SA} , producing disjoint epistemic outcome regions whose state-dependent weights are evaluated using the prepared epistemic density. In the finite-dimensional programme, the abstract region formalism developed here is combined with the Born-weight result of Paper B, while the full generalised-measurement and sequential-update layers are completed in TN4 and TN6 respectively. Decoherence is modelled through geometric separation on Σ of environmental degrees of freedom, with pointer states identified by stability under Fubini–Study volume spreading.

Section 6 introduces additional quantum structures, including interference as symplectic area and contextuality as geometric incompatibility of partitions, with examples based on low-dimensional projective manifolds.

Technical proofs, extended derivations, and auxiliary results are collected in the appendices.

Before developing the ontic and epistemic layers in Section 2, Table 1 summarises the geometric and measure-theoretic notation used throughout the paper.

This paper forms the third component of a sequence of developments in Constraint Surface Dynamics (CSD):

1. **Paper A** established volume typicality on Σ and showed how empirical frequencies arise from deterministic Hamiltonian flow.
2. **Paper B** proved the uniqueness of the Fubini–Study measure on $\mathbb{CP}^{n-1}$ under $SU(n)$ invariance.
3. **Paper C** (the present work) reconstructs finite-dimensional quantum mechanics by combining deterministic dynamics on Σ , an epistemic projection π , and the geometry of Fubini–Study outcome regions.
4. **Paper D** develops the interpretational layer, macroscopic irreversibility, and the phenomenology of measurement.
5. **Paper E** (forthcoming) will treat relativistic structure, identical particles, and continuous- and field-theoretic limits.

This work can be read independently as a self-contained reconstruction of finite-dimensional non-relativistic quantum mechanics.

2 Ontic Layer: The Configuration Surface Σ

The ontic description of a finite-dimensional quantum system is encoded in a compact Kähler manifold $(\Sigma, \omega_\Sigma, J_\Sigma, g_\Sigma)$ equipped with a deterministic Hamiltonian flow. This section specifies the geometric and measure-theoretic structure of Σ , formulates observables as smooth real functions, and explains how the epistemic Hilbert space geometry emerges strictly through the projection map $\pi : \Sigma \rightarrow \mathbb{CP}^{n-1}$.

Table 1: Notation used throughout this paper.

Symbol	Meaning
Σ	Ontic configuration manifold (finite-dimensional compact Kähler manifold)
ω_Σ	Symplectic 2-form on Σ
J_Σ	Complex structure on Σ compatible with ω_Σ
g_Σ	Riemannian metric on Σ compatible with $(\omega_\Sigma, J_\Sigma)$
μ_L	Liouville measure on Σ , $\mu_L = \omega_\Sigma^d / d!$ for $\dim_{\mathbb{R}} \Sigma = 2d$
$x \in \Sigma$	Ontic microstate of the system
$\omega(t)$	Ontic trajectory in Σ generated by the Hamiltonian flow
Φ_t	Hamiltonian flow on Σ , $\frac{d}{dt} \Phi_t(x) = X_H(\Phi_t(x))$
$\mathbb{C}^n$	Hilbert space for an n -level quantum system
$\mathbb{CP}^{n-1}$	Complex projective space of rays $[\psi]$ in $\mathbb{C}^n$
$\psi \in \mathbb{C}^n$	Hilbert vector representative of a ray
$[\psi] \in \mathbb{CP}^{n-1}$	Epistemic state (ray) corresponding to ψ
ω_{FS}	Fubini–Study symplectic form on $\mathbb{CP}^{n-1}$
g_{FS}	Fubini–Study Riemannian metric on $\mathbb{CP}^{n-1}$
μ_{FS}	Fubini–Study volume measure on $\mathbb{CP}^{n-1}$
d_{FS}	Fubini–Study distance on $\mathbb{CP}^{n-1}$
$\pi : \Sigma \rightarrow \mathbb{CP}^{n-1}$	Epistemic projection from ontic states to rays
π_*	Pushforward of measures and vector fields under π
ρ	Epistemic density on $\mathbb{CP}^{n-1}$, defined by $\pi_* \mu_L = \rho \mu_{\text{FS}}$
$H : \Sigma \rightarrow \mathbb{R}$	Ontic Hamiltonian generating deterministic dynamics on Σ
X_H	Hamiltonian vector field on Σ associated with H
$\hat{H}$	Self-adjoint Hamiltonian operator on $\mathbb{C}^n$ generating Schrödinger evolution
$\hbar$	Reduced Planck constant
$\Omega_i(M)$	Epistemic outcome region for outcome i of measurement M
Σ_{AB}	Joint ontic manifold for a bipartite system A and B
Σ_A, Σ_B	Ontic manifolds of subsystems A and B
π_{AB}	Joint projection $\Sigma_{AB} \rightarrow \mathbb{CP}^{n_{AB}-1}$
π_A, π_B	Marginal projections obtained from π_{AB}
ρ_{AB}	Joint epistemic density on $\mathbb{CP}^{n_{AB}-1}$
ρ_A, ρ_B	Reduced epistemic descriptions obtained by marginalisation of ρ_{AB}

2.1 Structural Definition of Σ

Axiom A1 states that Σ is a connected, finite-dimensional compact Kähler manifold

$$(\Sigma, \omega_\Sigma, J_\Sigma, g_\Sigma),$$

with

$$g_\Sigma(u, v) = \omega_\Sigma(u, J_\Sigma v).$$

Compactness ensures finiteness of the Liouville measure

$$\mu_L = \frac{\omega_\Sigma^d}{d!}, \quad d = \frac{1}{2} \dim \Sigma.$$

Axiom A4 introduces a smooth, many-to-one epistemic projection

$$\pi : \Sigma \longrightarrow \mathbb{CP}^{n-1},$$

whose fibres $\mathcal{F}_{[\psi]} = \pi^{-1}([\psi])$ encode ontic information invisible at the epistemic level. Paper B (FS uniqueness theorem) and Appendix G together guarantee that the pushforward measure satisfies

$$\pi_* \mu_L = \rho \mu_{\text{FS}}, \quad \rho \in L^1(\mathbb{CP}^{n-1}), \quad \rho > 0$$

and that μ_{FS} is the *unique* $\text{SU}(n)$ -invariant Kähler measure compatible with the induced geometry. Importantly, no Hilbert space structure is assumed ontically; the appearance of projective Hilbert geometry is wholly epistemic.

2.2 Observables as Smooth Functions

In this framework an observable corresponds to a smooth function on the epistemic space rather than to an operator introduced at the ontic level. Let $h_f: \mathbb{CP}^{n-1} \rightarrow \mathbb{R}$ be the smooth function associated with an observable f . In the quantum-effective sector described in Section 3, such functions may arise from Hermitian operators through the quadratic expectation-value map, but this representation is not assumed at the ontic level and is not required for the reconstruction.

The ontic manifold Σ carries the deterministic Hamiltonian flow generated by H . Since $\pi: \Sigma \rightarrow \mathbb{CP}^{n-1}$ is many-to-one, a single epistemic state $[\psi]$ corresponds to an entire fibre $\pi^{-1}([\psi])$ of ontic states. The additional fibre coordinates do not represent hidden variables that assign pre-existing values to observables. Their role is geometric rather than operational. They encode the internal structure needed to support Liouville-preserving flow on Σ , the absolute continuity condition $\pi_*\mu_L \ll \mu_{\text{FS}}$, and the context-dependent partitions that measurement interactions induce on $\mathbb{CP}^{n-1}$. The fibre degrees of freedom are therefore not measurable and do not determine values for incompatible observables.

Expectation values are defined at the epistemic level. Given an epistemic probability density ρ_{ep} with respect to the Fubini–Study measure μ_{FS} , the expectation of an observable f is

$$\langle f \rangle = \int_{\mathbb{CP}^{n-1}} h_f \rho_{\text{ep}} d\mu_{\text{FS}}.$$

This expression depends only on the epistemic state, not on the fibre coordinates of the ontic states in $\pi^{-1}([\psi])$. In particular, deterministic evolution of $\omega(t) \in \Sigma$ does not imply noncontextual value definiteness on $\mathbb{CP}^{n-1}$, a point that ensures consistency with the Kochen–Specker, Bell, and Fine constraints.

The dynamical evolution of observables follows from the projected flow on $\mathbb{CP}^{n-1}$. If H lies in the quantum-effective sector, then the induced projected vector field generates a Hamiltonian flow on $\mathbb{CP}^{n-1}$ up to controlled corrections of order $O(\epsilon)$. In the ideal limit $\epsilon = 0$, this reduces to the exactly projectable case.

Thus observables in this reconstruction are smooth epistemic functions whose values and dynamics are determined by the geometry of $\mathbb{CP}^{n-1}$ together with the projected Hamiltonian flow. No additional ontic observables or noncontextual value assignments are introduced. The observable algebra arises at the epistemic level from the structure inherited through the projection π and the symplectic geometry of projective space.

2.3 Emergent Hilbert Structure as an Epistemic Coordinate Chart

The projection π maps Σ onto the complex projective manifold $\mathbb{CP}^{n-1}$, which carries a canonical Kähler structure independent of any ontic Hilbert space. A representative $\psi \in \mathbb{C}^n$ of an epistemic class $[\psi]$ provides local coordinates; however, ψ is not a physical wavefunction, but an epistemic label for regions of Σ .

The Fubini–Study metric on $\mathbb{CP}^{n-1}$ is

$$g_{\text{FS}}([\phi], [\psi]) = \arccos^2 \left(\frac{|\langle \phi, \psi \rangle|}{\|\phi\| \|\psi\|} \right),$$

and arises entirely from the projective Kähler geometry. The inner product $\langle \phi, \psi \rangle$ is therefore an *epistemic* construction: it expresses geometric relations between equivalence classes under π , not ontic amplitudes.

Although $\mathbb{CP}^{n-1}$ is globally curved, any two points determine a unique complex projective geodesic. Local linear structure therefore emerges from geodesic interpolation, providing a geometric analogue of superposition without assuming linearity in Σ .

Quadratic Hamiltonians on $\mathbb{CP}^{n-1}$,

$$h([\psi]) = \frac{\langle \psi, \hat{H} \psi \rangle}{\langle \psi, \psi \rangle},$$

correspond to Hermitian operators $\hat{H}$ (Appendix I). Thus the operator formalism of standard quantum mechanics is recovered as an *epistemic* representation of geometric flows derived from Σ .

2.4 Integration with the Geometric Probability Framework

Appendix D studies a finite-dimensional projective exemplar in which the outcome regions $\Omega_i(M)$ are realised by Voronoi partitions of $\mathbb{CP}^{n-1}$ with respect to the FS metric. The abstract definition of outcome regions is given instead by Axiom A7, which requires only context-dependent measurable partitions with μ_{FS} -null boundaries.

Paper A establishes that long-run outcome frequencies are determined by invariant measure, while Paper B identifies μ_{FS} as the canonical epistemic measure on CP^{n-1} . Accordingly, for a prepared epistemic density ρ^{ep} ,

$$P(i \mid \rho^{ep}) = \int_{\Omega_i} \rho^{ep} d\mu_{FS}.$$

and Paper B shows that μ_{FS} is the *unique* measure compatible with $SU(n)$ -invariance. Thus CSD reproduces quantum measurement statistics through the combination of:

- deterministic Liouville–preserving Hamiltonian flow on Σ ,
- epistemic projection π ,
- FS geometry of CP^{n-1} ,
- and the unique invariant measure μ_{FS} .

The ontic structure of Σ therefore supports both the kinematic geometry of quantum states and the deterministic underpinning of Born-form probabilities, without introducing stochasticity at the fundamental level.

3 Epistemic Layer: Dynamics on CP^{n-1}

The epistemic dynamics of a finite-dimensional quantum system arises as the projection of deterministic Hamiltonian flow on the ontic manifold Σ into the complex projective manifold CP^{n-1} . This section formulates the induced vector fields, establishes conditions under which the evolution coincides with Schrödinger dynamics, and highlights the geometric origin of commutators, uncertainty relations, and Born weights.

3.1 Hamiltonian Flow on Σ

The deterministic evolution of the ontic microstate is described by a Hamiltonian flow on the compact Kähler manifold Σ . The manifold carries a symplectic form ω_Σ , a complex structure J_Σ , and the associated Riemannian metric g_Σ . These structures form part of the axioms of the reconstruction and provide the geometric framework for ontic dynamics. A smooth Hamiltonian function $H: \Sigma \rightarrow \mathbb{R}$ determines a unique Hamiltonian vector field X_H through

$$\iota_{X_H} \omega_\Sigma = dH,$$

which defines the flow Φ_t by

$$\frac{d}{dt} \omega(t) = X_H(\omega(t)), \quad \omega(0) \in \Sigma.$$

Since Σ is compact and H is smooth, the flow exists for all real times and each initial point specifies a unique trajectory.

The Hamiltonian flow preserves the Liouville measure μ_L on Σ . If U is any measurable region of Σ , then

$$\mu_L(\Phi_t(U)) = \mu_L(U)$$

for all t . This follows from the identity $\mathcal{L}_{X_H} \omega_\Sigma = 0$, which implies $\mathcal{L}_{X_H} \omega_\Sigma^d = 0$ for $2d = \dim \Sigma$ and ensures preservation of the symplectic volume form. Volume conservation on Σ is a key structural feature that supports the emergence of probabilistic behaviour in the epistemic description.

The Hamiltonian may describe an isolated system or a subsystem of a larger composite. In every case the ontic trajectory $\omega(t)$ is determined entirely by the Hamiltonian and the initial condition, without any stochastic element. The epistemic dynamics considered in the following subsections arise by projecting this deterministic flow through the map $\pi: \Sigma \rightarrow CP^{n-1}$, which associates each ontic state with an epistemic ray. The projection introduces a coarse-graining from ontic to epistemic space while leaving the deterministic character of the evolution intact.

3.2 Projection to Epistemic Space

The epistemic description of the system is obtained by projecting the ontic manifold Σ onto the complex projective space CP^{n-1} . The projection map

$$\pi: \Sigma \rightarrow CP^{n-1}$$

associates each ontic microstate with an epistemic ray. Since Σ generally has higher dimension than CP^{n-1} the map π is many to one. The projection therefore groups many ontic microstates into the same epistemic equivalence class. The epistemic state $[\psi] = \pi(\omega)$ represents the operationally accessible description of the system, while the ontic state

Figure 1: The projection $\pi : \Sigma \rightarrow \mathbb{CP}^{n-1}$ collapses each fibre to an epistemic ray. Deterministic dynamics on Σ project to effective Hamiltonian flows on $\mathbb{CP}^{n-1}$ within the quantum-effective sector, and become exactly projectable in the ideal fibre-invariant limit.

contains additional structure that is not visible at the epistemic level. The projection does not introduce any stochastic element and the evolution remains deterministic.

The deterministic Hamiltonian flow on Σ induces a flow on epistemic space whenever the pushforward vector field $\pi_* X_H$ is well defined. For any smooth function h on $\mathbb{CP}^{n-1}$ the derivative along the projected trajectory satisfies

$$\frac{d}{dt} h(\pi(\omega(t))) = dh_{\pi(\omega(t))}(\pi_* X_H).$$

A sufficient exact condition for $\pi_* X_H$ to be well defined is that the Hamiltonian be constant along the fibres of π , so that $H = h \circ \pi$ for some smooth h on $\mathbb{CP}^{n-1}$. More generally, as shown in Appendix I, a consistent projected evolution also exists as an effective description whenever the fibre dependence is uniformly small in the sense that $H = h \circ \pi + \delta H$ with $\|d(\delta H)|_V\| = O(\epsilon)$ over the operational interval of interest. The quantum-effective sector studied below is defined by this controlled approximate projectability.

The projected space $\mathbb{CP}^{n-1}$ carries the Fubini–Study metric and the associated measure μ_{FS} . As shown in Paper B this measure is uniquely fixed by $\text{SU}(n)$ symmetry. It plays a central role in the statistical reconstruction developed in Sections 5 and 6. The projection π therefore provides the link between deterministic ontic evolution and the geometry of epistemic space, which forms the basis for the derivation of Schrödinger dynamics in Section 3.3.

3.3 Non-linear flows and the quantum-effective sector

The deterministic Hamiltonian flows that exist on the ontic manifold Σ are generally more numerous than those that project to the linear evolution of quantum mechanics. A Hamiltonian on Σ need not satisfy the exact fibre-invariance condition, nor even lie in the approximately projectable regime described in Appendix I. Outside the quantum-effective sector, the projected dynamics on $\mathbb{CP}^{n-1}$ need not be well defined even approximately, or may fail to preserve the Kähler structure to the accuracy required for a quantum description. The resulting projected dynamics are typically non-linear and need not coincide with the unitary evolution of standard quantum theory. These flows represent mathematically admissible deterministic dynamics on Σ but are not assumed to be physically realised.

In laboratory quantum regimes, it is natural to expect the fibre-sensitive contribution to the ontic Hamiltonian to be effectively suppressed by timescale separation, weak coupling, or experimental coarse-graining, so that $H = h \circ \pi + \delta H$ with $\|d(\delta H)|_V\| = O(\epsilon)$ and $\epsilon \ll 1$ over the operational interval of interest.

The quantum-effective Hamiltonians are those for which the ontic Hamiltonian admits a decomposition

$$H = h \circ \pi + \delta H,$$

with fibre variation bounded by

$$\|d(\delta H)|_V\| = O(\epsilon), \quad \epsilon \ll 1,$$

throughout the dynamically relevant sector. In the ideal limit $\epsilon = 0$, this reduces to exact fibre-invariance.

In this regime the projected vector field is well defined up to controlled corrections of order $O(\epsilon)$ and preserves the symplectic and complex structure of the Fubini–Study manifold to the same accuracy over finite operational times. Schrödinger evolution is therefore recovered as the leading effective epistemic dynamics associated with deterministic ontic flow in this sector.

Axiom A5 should be read as a sector-selection postulate for the finite-dimensional quantum-effective regime, not as a derived theorem about all ontic Hamiltonians on Σ . The claim established here is conditional: if the ontic dynamics lies in the approximately projectable class specified by A5, then the projected epistemic dynamics closes and takes Schrödinger form up to the controlled error bounds stated in Appendix I. A deeper physical derivation of why physically realised laboratory Hamiltonians should lie in this sector is not provided in the present paper.

3.4 Projection to Schrödinger Evolution

The previous subsection identified the quantum-effective sector as the class of Hamiltonians on Σ for which the fibre dependence is perturbatively small, so that

$$H = h \circ \pi + \delta H, \quad \|d(\delta H)|_V\| = O(\epsilon),$$

with $\epsilon \ll 1$ on the operational timescale of interest. In the exact limit $\epsilon = 0$, one recovers fibre-invariance and an exactly well-defined projected flow.

For such Hamiltonians the projected vector field is well defined up to controlled corrections of order $O(\epsilon)$ and preserves the Kähler structure of the epistemic space to leading order over finite times.

A theorem of Ashtekar and Schilling (Ashtekar and Schilling (1999)) shows that the holomorphic Hamiltonian vector fields on $\mathbb{CP}^{n-1}$ correspond precisely to flows generated by quadratic Hamiltonians of the form

$$h([\psi]) = \frac{\langle \psi, \hat{H} \psi \rangle}{\langle \psi, \psi \rangle}, \quad \hat{H} = \hat{H}^\dagger.$$

In any homogeneous coordinate chart the associated epistemic dynamics satisfy

$$\frac{d}{dt} |\psi(t)\rangle = -\frac{i}{\hbar} \hat{H} |\psi(t)\rangle,$$

where $|\psi(t)\rangle$ is a representative of the evolving ray $[\psi(t)]$. This is the Schrödinger equation. The wavefunction is not introduced at the ontic level. Instead, it appears as a coordinate representation of the epistemic state and inherits its linearity from the geometric properties of the projected flow.

The Schrödinger equation therefore arises within this reconstruction as the leading effective epistemic description of deterministic flows on the ontic manifold whose fibre-sensitive corrections remain perturbatively small while the projected dynamics preserves the symplectic and complex structures of projective Hilbert space to the required accuracy.

3.5 Poisson Brackets and Commutators

The ontic manifold Σ carries a natural Poisson structure defined by

$$\{f, g\}_\Sigma = \omega_\Sigma(X_f, X_g),$$

for all smooth functions f and g on Σ , where X_f and X_g are the Hamiltonian vector fields associated with these functions. For Hamiltonians in the quantum-effective sector, the projection $\pi : \Sigma \rightarrow \mathbb{CP}^{n-1}$ induces a Poisson structure on epistemic space up to controlled corrections of order $O(\epsilon)$. In the ideal limit $\epsilon = 0$, this reduces to the exactly projectable case $H = h \circ \pi$.

If h_1 and h_2 arise as the effective projected Hamiltonians associated with ontic Hamiltonians in the quantum-effective sector, their Poisson bracket on projective space is

$$\{h_1, h_2\}_{\text{FS}} = \omega_{\text{FS}}(X_{h_1}, X_{h_2}),$$

where ω_{FS} is the Fubini–Study symplectic form. The projected flows preserve the Kähler structure of $\mathbb{CP}^{n-1}$ and therefore define holomorphic Hamiltonian vector fields.

When h_1 and h_2 are quadratic Hamiltonians associated with Hermitian operators $\hat{H}_1$ and $\hat{H}_2$,

$$h_i([\psi]) = \frac{\langle \psi, \hat{H}_i \psi \rangle}{\langle \psi, \psi \rangle},$$

the Fubini–Study Poisson bracket reproduces the operator commutator. A standard calculation in any homogeneous coordinate chart gives

$$\{h_1, h_2\}_{\text{FS}} = \frac{1}{i\hbar} \frac{\langle \psi, [\hat{H}_1, \hat{H}_2] \psi \rangle}{\langle \psi, \psi \rangle}.$$

The commutator algebra of quantum observables therefore emerges as the epistemic image of the Poisson geometry of Σ . No operator structure is assumed at the ontic level. The algebra of observables appears naturally from the symplectic properties of the projective space on which epistemic states evolve.

3.6 Geometric Uncertainty Relations

Observable quantities in the epistemic description are represented by smooth functions h_f on $\mathbb{CP}^{n-1}$ arising from Hamiltonians in the quantum-effective sector of Σ . In the exact limit $\epsilon = 0$, these reduce to functions induced by fibre-invariant Hamiltonians. Given an epistemic probability density ρ_{ep} with respect to the Fubini–Study measure μ_{FS} , the expectation value of h_f is

$$\langle h_f \rangle = \int_{\mathbb{CP}^{n-1}} h_f \rho_{\text{ep}} d\mu_{\text{FS}},$$

and the variance is defined by

$$(\Delta f)^2 = \int_{\mathbb{CP}^{n-1}} (h_f - \langle h_f \rangle)^2 \rho_{\text{ep}} d\mu_{\text{FS}}.$$

The Fubini–Study metric and symplectic form relate these statistical quantities to the geometry of $\mathbb{CP}^{n-1}$. For observables f and g with Hamiltonian functions h_f and h_g , the Fubini–Study Poisson bracket is

$$\{h_f, h_g\}_{\text{FS}} = \omega_{\text{FS}}(X_{h_f}, X_{h_g}).$$

Applying the Cauchy–Schwarz inequality to the Fubini–Study metric and using the compatibility of the metric with the symplectic structure yields the geometric uncertainty relation

$$\Delta f \Delta g \geq \frac{1}{2} |\langle \{h_f, h_g\}_{\text{FS}} \rangle|.$$

This inequality depends only on the intrinsic geometry of the projective state space and does not require any operator formalism.

For quadratic Hamiltonians corresponding to Hermitian operators $\hat{F}$ and $\hat{G}$ the Poisson bracket reduces to

$$\{h_f, h_g\}_{\text{FS}} = \frac{1}{i\hbar} \frac{\langle \psi, [\hat{F}, \hat{G}] \psi \rangle}{\langle \psi, \psi \rangle},$$

so the geometric uncertainty relation reproduces the standard Robertson relation. Within this reconstruction the uncertainty principle arises directly from the symplectic and metric geometry of $\mathbb{CP}^{n-1}$ and does not rely on additional postulates.

3.7 Born-form probabilities from epistemic measure and operational consistency

The projection map $\pi : \Sigma \rightarrow \mathbb{CP}^{n-1}$ and the Fubini–Study measure μ_{FS} allow outcome probabilities to be represented on the epistemic state space. For a measurement context M , let $\{\Omega_i(M)\} \subset \mathbb{CP}^{n-1}$ be measurable outcome regions with μ_{FS} -null boundaries. In the projective examples studied here, these regions admit explicit Voronoi realisations, but the probability rule does not depend on that special construction.

Given a prepared epistemic density ρ^{ep} with respect to μ_{FS} , define

$$P(i \mid M, \rho^{\text{ep}}) = \int_{\Omega_i(M)} \rho^{\text{ep}} d\mu_{\text{FS}}.$$

For a sharply prepared pure state, the canonical idealisation is the rank-one operator

$$\hat{\rho}_{[\psi]} = |\psi\rangle\langle\psi|,$$

with probabilities

$$P(i \mid \hat{\rho}_{[\psi]}, M) = \text{Tr}(\hat{\rho}_{[\psi]} \Pi_i(M)).$$

Within the CSD programme, Paper B establishes that the admissible probability functional on projective measurement events is fixed by symmetry and operational consistency, while Paper A explains how deterministic volume-typicality on Σ yields the corresponding long-run frequencies. The role of the Fubini–Study geometry here is to provide the canonical invariant epistemic measure and a natural measurable realisation of outcome regions, not to identify generic sharp-state probabilities with raw neighbourhood-volume intersections in $\mathbb{CP}^{n-1}$.

4 Composite Systems

Composite quantum systems require special treatment within the CSD framework because the ontic configuration surface does not factorise into subsystem components, even when the epistemic description approximately does so. This section clarifies the geometric structure of Σ_{AB} , the emergence of entanglement from Hamiltonian coupling, and the construction of reduced epistemic states.

4.1 Ontic Composite Structure

Axiom A6 states that the ontic configuration space of a bipartite system is a single connected Kähler manifold

$$\Sigma_{AB},$$

and *does not* decompose into a Cartesian product:

$$\Sigma_{AB} \neq \Sigma_A \times \Sigma_B.$$

The degrees of freedom belonging to A and B may be locally identifiable but are not ontically independent.

The epistemic projection

$$\pi_{AB} : \Sigma_{AB} \longrightarrow \mathbb{CP}^{n_{AB}-1}$$

admits subsystem marginals (Section 3), and in regimes of weak or negligible interaction the image of Σ_{AB} under π_{AB} approximates the product structure

$$\pi_{AB}(\Sigma_{AB}) \approx \mathbb{CP}^{n_A-1} \times \mathbb{CP}^{n_B-1}$$

on sufficiently small neighbourhoods. This factorisation is therefore an *epistemic* approximation rather than an ontic identity. Σ_{AB} is intrinsically nonfactorising, whereas the epistemic space may locally appear factorised.

4.2 Non-factorisation of the ontic space

The composite ontic manifold Σ_{AB} introduced in the previous subsection is not required to factorise as a Cartesian product $\Sigma_A \times \Sigma_B$, even in the case where the interaction Hamiltonian H_{int} vanishes. This reflects the possibility that Σ_{AB} carries global geometric structure that does not admit a description in terms of independent subsystem coordinates. Such structure allows the ontic manifold to encode correlations that become visible only after projection through π_{AB} and that correspond to entangled epistemic states. These correlations are features of the geometry of Σ_{AB} and do not rely on the presence of a dynamical coupling, so they need not disappear when $H_{\text{int}} = 0$. The non-factorising structure of the ontic manifold is therefore an essential part of the reconstruction rather than an optional complication.

Although Σ_{AB} need not factorise globally, an approximate factorisation arises whenever the system is prepared in a product epistemic state $[\psi_A] \otimes [\psi_B]$ and the interaction term is negligible. In this regime the projection map behaves locally as

$$\pi_{AB}(x_{AB}) \approx (\pi_A(x_A), \pi_B(x_B))$$

for all ontic microstates in the support of the preparation. This recovers the expected product structure of subsystem epistemic descriptions and ensures that separable states propagate correctly. The approximate factorisation breaks down precisely in the regimes where quantum mechanics describes entanglement, matching the empirical domain of non-separable correlations.

The reduced epistemic states of the subsystems are obtained by marginalisation of the joint pushforward density ρ_{AB} through

$$\rho_A = (\pi_A)_* \rho_{AB}, \quad \rho_B = (\pi_B)_* \rho_{AB}.$$

In the quadratic (unitary) sector, this marginalisation is intended to coincide with the standard reduced-state construction that reproduces subsystem measurement statistics. Establishing full equivalence with the operator partial trace for arbitrary joint preparations is deferred.

The fact that Σ_{AB} does not factorise does not imply any violation of relativistic causality. The reconstruction enforces that subsystem epistemic descriptions factorise whenever the operational conditions for separability hold, and no signalling channel is introduced by the geometric correlations that reside at the ontic level.

4.3 Interaction and Entanglement

Let the total Hamiltonian be

$$H = H_A + H_B + H_{\text{int}},$$

where H_{int} generates coupling between the subsystems. The ontic flow on Σ_{AB} is

$$\frac{dx}{dt} = X_H(x), \quad x \in \Sigma_{AB}.$$

Within the quantum-effective sector defined in Appendix I, the projected vector field on the relevant epistemic space is well defined up to controlled corrections of order $O(\epsilon)$. Accordingly, the composite-system dynamics discussed here

should be understood as the leading effective projected dynamics, with exact projectability recovered only in the ideal limit $\epsilon = 0$.

Thus entanglement emerges as an epistemic phenomenon associated with the geometry of $\text{CP}^{n_{AB}-1}$ and the nonfactorising structure of Σ_{AB} .

If H_{int} is sufficiently weak or acts only over short timescales, the flow remains close to the product manifold

$$\text{CP}^{n_A-1} \times \text{CP}^{n_B-1},$$

but only as an approximation. CSD therefore distinguishes:

- ontic nonfactorisation (exact),
- epistemic approximate factorisation (regime dependent).

4.4 Subsystem Marginals and Reduced States

The marginal epistemic map

$$\pi_A : \Sigma_{AB} \longrightarrow \text{CP}^{n_A-1}$$

is defined by integrating out the degrees of freedom associated with subsystem B . Let ρ_{AB} be an epistemic density on $\text{CP}^{n_{AB}-1}$, induced by a density on Σ_{AB} . The marginal is defined by

$$\rho_A = (\pi_A)_* \rho_{AB}.$$

Given the fibre decomposition

$$\pi_{AB}^{-1}([\psi_{AB}]) \rightarrow \pi_A^{-1}([\psi_A]),$$

the marginal density is obtained by integrating ρ_{AB} over the fibres corresponding to subsystem B . This is the geometric analogue of the partial trace.

Equivalently, Appendix F shows that for nondegenerate measurements on subsystem A , marginalisation can be expressed as

$$\rho_A([\psi_A]) = \int_{\text{SU}(n_B)} \rho_{AB}(U_B \cdot [\psi_A \otimes \phi_B]) dU_B,$$

which is consistent with the standard reduced-state construction in the quadratic (unitary) sector; a general equivalence-to-partial-trace proof is deferred.

Thus reduced states and mixed states arise in CSD purely from pushforward measures on CP^{n-1} , with no ontic stochasticity.

4.5 Entanglement Geometry

For a pair of qubits, the epistemic state space is

$$\text{CP}^3,$$

within which the separable sector is a complex quadric

$$\mathcal{S} \subset \text{CP}^3.$$

The set $\mathcal{S}$ is a measure-zero submanifold of CP^3 under μ_{FS} , while entangled states occupy the complement. The nonfactorising structure of Σ_{AB} implies that typical ontic trajectories project into entangled regions even when initial conditions are nearly separable.

Because Σ_{AB} does not admit a product decomposition, entanglement does not reflect any ontic “joining” of subsystem states. Instead, it reflects the geometry of the epistemic manifold and the pushforward of deterministic flow into a space where tensor product structure is emergent rather than fundamental. CSD therefore supports the standard quantum distinction between separable and entangled states while grounding it in deterministic ontic dynamics.

5 Measurement, Decoherence, and Pointer Basis

Measurement in CSD is implemented as deterministic Hamiltonian evolution on a joint ontic surface describing the system, apparatus, and, when relevant, the environment. No stochastic ingredients are introduced at the ontic level; apparent branching and outcome statistics arise from geometric separation in Σ and from the epistemic projection into CP^{n-1} . This section develops the measurement model, defines outcome regions, describes decoherence as geometric isolation, and explains the emergence of pointer bases.

5.1 Measurement as deterministic flow

Measurement is treated in this reconstruction as a fully deterministic physical process. A system with ontic state $\omega \in \Sigma_S$ interacts with an apparatus whose ontic state lies in Σ_A , giving a composite ontic space Σ_{SA} . The joint evolution is generated by a Hamiltonian of the form

$$H_{SA} = H_S + H_A + H_{\text{int}},$$

and the composite trajectory $\omega(t)$ in Σ_{SA} is determined uniquely by the Hamiltonian vector field $X_{H_{SA}}$.

The epistemic description is obtained by projection through the system map $\pi_S: \Sigma_{SA} \rightarrow \mathbb{CP}^{n-1}$. The apparatus degrees of freedom determine a preferred set of pointer rays $\{[\phi_i]\}$, and these rays induce a geometric partition of epistemic space. In the projective model studied here, the outcome regions are realised as Fubini–Study Voronoi cells, as described in Appendix D

$$\Omega_i(M) \subset \mathbb{CP}^{n-1}$$

constructed from the set $\{[\phi_i]\}$ associated with the measurement context M . The regions are therefore defined entirely on epistemic space and do not require any partition of the ontic manifold Σ_{SA} .

The rule for determining an outcome is non-circular. The partition $\{\Omega_i(M)\}$ is fixed by the measurement context through the pointer structure of the apparatus, independently of the ontic state. A definite result is selected by the ontic trajectory through its epistemic image. Once the interaction has produced a stable separation of pointer sectors, a recorded outcome i is defined by

$$\pi_S(\omega(t)) \in \Omega_i(M),$$

for any t greater than the duration of the interaction. The determinism of this rule follows from the fact that the boundaries of the Voronoi regions have Fubini–Study measure zero, as shown in Appendix G. Deterministic trajectories almost surely never intersect region boundaries, so the assignment of outcomes is unambiguous.

The dimensional mismatch between Σ_{SA} and $\mathbb{CP}^{n-1}$ poses no difficulty. Many ontic microstates may project to the same epistemic ray, but the realised result is determined solely by the epistemic location of $\pi_S(\omega(t))$ within the partition. Measurement in this reconstruction is therefore deterministic Hamiltonian evolution on Σ_{SA} together with epistemic coarse-graining under the projection π_S . The apparatus defines the epistemic partition, while the ontic trajectory determines which region is reached, yielding a clear and non-circular account of measurement.

5.2 Outcome Regions Ω_i

A measurement context M corresponds to a set of apparatus–distinguished rays

$$\{[\phi_i]\} \subset \mathbb{CP}^{n-1}.$$

In the projective examples studied here, the corresponding outcome regions are realised using FS Voronoi geometry, as described in Appendix D:

$$\Omega_i(M) = \{[\psi] : d_{\text{FS}}([\psi], [\phi_i]) < d_{\text{FS}}([\psi], [\phi_j]), j \neq i\}.$$

Appendix G shows that the boundaries

$$\partial\Omega_i = \{[\psi] : d_{\text{FS}}([\psi], [\phi_i]) = d_{\text{FS}}([\psi], [\phi_j])\}$$

are smooth hypersurfaces of codimension one and have μ_{FS} -measure zero. Thus the measurable event structure is defined geometrically through the Fubini–Study partition, while outcome probabilities are assigned by the prepared epistemic density relative to that partition:

$$P(i \mid \rho^{ep}) = \int_{\Omega_i} \rho^{ep} d\mu_{\text{FS}}.$$

In realistic apparatus models (Appendix B), the mapping from initial states to final asymptotic rays is determined by H_{int} . The set of rays $\{[\phi_i]\}$ corresponds to stable asymptotic directions of flow, providing a purely geometric interpretation of measurement outcomes.

Figure 2: Fubini–Study Voronoi partition of $\mathbb{CP}^1 \cong S^2$ for a two-outcome measurement. The pointer states define the poles and the outcome regions are spherical caps separated by a geodesic of μ_{FS} -measure zero.

5.3 Pointer Basis

The interaction Hamiltonian H_{int} couples the system to the apparatus and generates a deterministic flow on the joint manifold Σ_{SA} . Its epistemic projection defines a flow on $\mathbb{CP}^{n_A-1}$ that separates into dynamically stable regions. Pointer states are identified with the rays that remain least distorted under this flow and therefore serve as the stable labels of the outcome sectors.

For a nondegenerate measurement the interaction selects a distinguished family of rays $\{[\phi_i]\}$ in the system space. Let $X_{H_{\text{int}}}$ be the projected Hamiltonian vector field on $\mathbb{CP}^{n-1}$ and define the local Fubini–Study dispersion functional by

$$\mathcal{D}_M([\psi]) = g_{\text{FS}}(X_{H_{\text{int}}}, X_{H_{\text{int}}})([\psi]).$$

The pointer states are the minimisers of this functional:

$$\mathcal{D}_M([\phi_i]) \leq \mathcal{D}_M([\psi]) \quad \text{for all } [\psi] \in \mathbb{CP}^{n-1}.$$

Their stability ensures that the projected state of the apparatus becomes correlated with the region $\Omega_i(M)$ into which the evolving system state is directed. These rays therefore define the centres of the Voronoi regions introduced in Appendix D.

Geometrically, pointer states correspond to the directions along which the interaction produces the smallest Fubini–Study expansion. Because the Voronoi boundaries have μ_{FS} -measure zero, these stability directions determine the macroscopic outcomes observed in the apparatus. No stochastic element or collapse mechanism is introduced. Pointer states arise from the structure of the interaction Hamiltonian and the geometry of $\mathbb{CP}^{n-1}$, and their robustness is consistent with the stability conditions found in decoherence theory while remaining fully deterministic at the level of Σ .

6 Higher Quantum Structure

Beyond single–outcome statistics and composite systems, quantum theory exhibits interference, contextuality, and higher–dimensional structure. In CSD these arise from geometric features of $\mathbb{CP}^{n-1}$ and its symplectic–Kähler structure, with no interpretational assumptions. This section summarises the reconstruction of these features within the finite–dimensional framework.

6.1 Interference and Symplectic Area

Two–path interference follows directly from symplectic geometry. Consider two epistemic paths $[\psi_1(t)]$ and $[\psi_2(t)]$ in $\mathbb{CP}^{n-1}$ joining the same endpoints. Define the closed loop γ by traversing one path forward and the other backward. The symplectic area enclosed by γ is

$$\mathcal{A}(\gamma) = \oint_{\gamma} \theta_{\text{FS}} = \int_S \omega_{\text{FS}},$$

where S is any surface bounded by γ , and θ_{FS} is the FS symplectic potential.

For two paths connecting rays $[\psi_a]$ and $[\psi_b]$, the Bargmann invariant is

$$\mathcal{B} = \langle \psi_a, \psi_1 \rangle \langle \psi_1, \psi_2 \rangle \langle \psi_2, \psi_a \rangle,$$

whose argument equals the geometric phase,

$$\arg \mathcal{B} = \frac{1}{\hbar} \mathcal{A}(\gamma).$$

Thus phase differences between paths arise purely from symplectic area in $\mathbb{CP}^{n-1}$, with interference fringes given by the relative area swept by the optically induced epistemic trajectories.

Because the dynamics on Σ is deterministic and Hamiltonian, interference reflects the geometry of the induced epistemic flow rather than superposition of ontic states.

6.2 Contextuality from Context-Dependent Partitions

In this framework contextuality arises from the incompatibility of context-dependent measurement partitions on the epistemic space CP^{n-1} . Each measurement context M generates a distinct family of outcome regions $\{\Omega_i(M)\}$, and different contexts generally induce different partitions because the interaction Hamiltonian selects different stability directions of the projected flow (see Section 5.5). In the projective examples studied here, these partitions are realised concretely by Fubini–Study Voronoi constructions. The resulting partitions are generally incompatible, meaning that their boundaries do not align and there is no common refinement that represents both simultaneously.

The deterministic microstate $\omega \in \Sigma$ does not assign values to all observables at once. The outcome revealed by a measurement depends on which region $\Omega_i(M)$ the epistemic state $[\psi] = \pi(\omega)$ occupies after the system has been coupled to the apparatus implementing context M . Changing the context changes the interaction Hamiltonian and therefore changes the outcome partition that defines outcomes. Determinism at the ontic level does not imply value-definiteness across incompatible contexts.

Because probabilities are computed from a prepared epistemic state ρ^{ep} or equivalently from $\hat{\rho}$, changing context changes the measurable event $\Omega_i(M)$ and the associated projector family $\{\Pi_i(M)\}$, not a context-independent raw volume assigned to an outcome label.

This resolves the Kochen–Specker obstruction. No global value map on CP^{n-1} is introduced, and no assumption of noncontextuality is made. The model is therefore consistent with the Kochen–Specker theorem and with the constraints of Bell and Fine. Contextual behaviour emerges from the fact that each measurement context induces its own coarse-graining of the epistemic manifold through its associated outcome partition, while the deterministic flow on Σ remains well defined throughout. In the projective examples studied here, these partitions admit explicit Fubini–Study Voronoi realisations.

Contextuality in this reconstruction is a geometric property of how different measurement interactions partition CP^{n-1} and not a sign of any inconsistency in the underlying ontic dynamics.

6.3 Qutrit Structure and CP^2

The state space of a qutrit is CP^2 , a four-dimensional compact Kähler manifold with $SU(3)$ symmetry. Its geometry is richer than that of CP^1 , containing nontrivial two-dimensional complex subspaces and curvature structures not present for qubits.

Key structures include:

- complex projective lines $CP^1 \subset CP^2$ corresponding to qubit subspaces;
- geodesic surfaces associated with two-level transitions;
- FS distance functions defining Voronoi outcome partitions for three-outcome measurements.

Appendix C develops the three-outcome Stern–Gerlach measurement as a concrete example, where the rays associated with spin-1 eigenstates define a three-cell FS Voronoi tessellation.

A complete treatment of subsystem embeddings, fibre geometry of π , and the constraints required for Lorentz and field-theoretic limits lies beyond the finite-dimensional scope of this paper. These issues are deferred to the Σ paper, where the global geometry of Σ and higher-dimensional field structures will be developed.

The present paper develops the general reconstruction for finite-dimensional systems but does not include explicit worked examples beyond the qubit case. Detailed illustrations of projective examples by Fubini–Study Voronoi partitions in higher-dimensional spaces such as CP^2 , together with explicit composite-system examples that show how π_{AB} and the induced marginals operate in practice, will appear in companion notes. These examples would extend the formalism presented here without altering the structure of the reconstruction itself.

7 Summary

This paper has reconstructed the full operational content of finite-dimensional quantum mechanics from the deterministic Hamiltonian flow of a single ontic state evolving on a compact Kähler manifold Σ . The results show that the statistical and dynamical structure of quantum theory arises from the geometry of the epistemic manifold CP^{n-1} and the unique invariant Fubini–Study measure, without assuming Hilbert space or probabilistic postulates at the ontic level.

The fundamental evolution on Σ is governed by a Hamiltonian vector field X_H preserving the Liouville measure. When the Hamiltonian lies in the quantum-effective sector defined in Axiom A5 and Appendix I, the epistemic projection π maps the ontic flow on Σ to a Hamiltonian flow on $\mathbb{CP}^{n-1}$ up to controlled corrections of order $O(\epsilon)$. If the corresponding effective projected Hamiltonian h is quadratic, this leading projected dynamics is the standard finite-dimensional Schrödinger evolution. Exact holomorphic unitary flow is recovered only in the ideal limit $\epsilon = 0$.

The state vector ψ is not an ontic object but an epistemic coordinate chart on $\mathbb{CP}^{n-1}$, representing an equivalence class of many ontic microstates. The physically real degrees of freedom reside on Σ , whose higher-dimensional structure contains additional information not captured in ψ . This geometric perspective clarifies that the Hilbert space formalism is a convenient epistemic representation rather than a fundamental ontology.

The geometric probability framework constructed in Papers A and B and used here shows that outcome probabilities are represented by integrals of the prepared epistemic density over context-dependent measurable outcome regions,

$$P(i) = \int_{\Omega_i} \rho^{ep} d\mu_{FS},$$

with the operator form $P(i | \hat{\rho}, M) = \text{Tr}(\hat{\rho}\Pi_i)$ used as the canonical sharp-state idealisation. In the projective examples studied here, the regions Ω_i admit explicit Voronoi realisations, but Voronoi geometry is illustrative rather than the general probability postulate.

Because μ_{FS} is the canonical $\text{SU}(n)$ -invariant epistemic measure, and because the admissible probability functional is fixed by the operational and symmetry constraints established in Paper B, the Born form is recovered without introducing stochastic ontic dynamics.

Measurement processes arise from deterministic coupling between system and apparatus on the joint ontic manifold Σ_{SA} , with apparent branching originating from the divergence of spatial or internal apparatus trajectories. Decoherence results from geometric isolation in Σ_{SAE} , where environmental degrees of freedom separate in phase space and induce effective diagonalisation in the epistemic layer. Composite systems are treated without assuming ontic factorisation: Σ_{AB} is a single connected manifold, yet its epistemic image approximates product structure when interactions are weak. Entanglement then emerges from Hamiltonian coupling, and reduced states arise from marginalisation via pushforward measures, reproducing the partial trace.

Contextuality appears as a geometric feature of incompatible context-dependent outcome partitions corresponding to different measurement contexts. In the projective examples studied here, this incompatibility is illustrated by the corresponding FS Voronoi realisations. This provides a clear operational account of contextual phenomena without invoking hidden variables or interpretive assumptions. Interference, geometric phase, and qutrit structure further demonstrate that the full phenomenology of finite-dimensional quantum mechanics is recovered from the underlying deterministic model.

Taken together, these results show that finite-dimensional quantum mechanics is fully reproduced by a deterministic flow on Σ combined with an epistemic projection into $\mathbb{CP}^{n-1}$. The unitary dynamics, Born probabilities, entanglement structure, decoherence behaviour, contextuality properties, and higher-level kinematics all emerge from the geometry of the epistemic space and the measure-preserving flow on the ontic manifold. Thus CSD provides a coherent geometric reconstruction of operational quantum theory, grounded in deterministic dynamics and without introducing stochasticity or nonlocal hidden variables.

A Axiom System for the Ontic Surface Σ

This appendix presents the axioms defining the ontic configuration manifold Σ and its relation to the epistemic manifold $\mathbb{CP}^{n-1}$. These axioms constitute the minimal geometric and dynamical assumptions required for the reconstruction in the main text. Supporting geometric, measure-theoretic, and dynamical results are developed in Appendices C, D, F, G, H, and I.

A.1 Axiom A1: Ontic Configuration Surface

The ontic state of a finite-dimensional quantum system is represented by a point x on a finite-dimensional compact Kähler manifold (Σ, ω, J, g) , where ω is the symplectic form, J is the complex structure, and g is the compatible Riemannian metric. Compactness ensures that the Liouville measure

$$\mu_L = \frac{\omega^n}{n!}$$

is finite. No relativistic structure, causal ordering, or spacetime foliation is assumed. The geometric framework follows standard finite-dimensional Kähler mechanics (Kibble 1979).

A.2 Axiom A2: Deterministic Hamiltonian Dynamics

The ontic state evolves under a deterministic Hamiltonian flow generated by a smooth Hamiltonian $H : \Sigma \rightarrow \mathbb{R}$. The associated vector field X_H satisfies

$$\mathcal{L}_{X_H}\omega = 0, \quad \mathcal{L}_{X_H}\mu_L = 0,$$

so that the Liouville measure is preserved. The evolution of states is given by $\Phi_t : \Sigma \rightarrow \Sigma$ with $\frac{d}{dt}\Phi_t(x) = X_H(\Phi_t(x))$.

A.3 Axiom A3: Epistemic Projection

Observers access only epistemic states represented by rays $[\psi] \in \mathbb{CP}^{n-1}$. There exists a smooth many-to-one projection

$$\pi : \Sigma \longrightarrow \mathbb{CP}^{n-1}$$

mapping ontic microstates to epistemic states. The fibres $\pi^{-1}([\psi])$ correspond to ontic distinctions that are not epistemically accessible. The existence of π is taken as a modelling assumption defining the epistemic layer. The geometric structure of $\mathbb{CP}^{n-1}$ and its coordinate charts is reviewed in Appendix C. The ontic–epistemic interpretation is developed conceptually in Paper D.

A.4 Axiom A4: Pushforward Measure and Absolute Continuity

The pushforward of the Liouville measure under π satisfies

$$\pi_*\mu_L = \rho d\mu_{\text{FS}},$$

where μ_{FS} is the Fubini–Study measure on $\mathbb{CP}^{n-1}$ and ρ is a smooth density. Absolute continuity holds:

$$\pi_*\mu_L \ll \mu_{\text{FS}}.$$

Paper B shows that invariance under complex rescaling and $\text{SU}(n)$ symmetry implies that ρ is constant, selecting μ_{FS} as the unique epistemic measure compatible with the reconstruction. The measure-theoretic justification, including regularity of Voronoi boundaries and absolute continuity properties, is presented in Appendix G.

A.5 Axiom A5: Quantum-Effective Hamiltonians

The finite-dimensional reconstruction developed in this paper does not require exact fibre-invariance of the ontic Hamiltonian on all of Σ . Instead, we assume that the experimentally relevant regime lies in a *quantum-effective sector* in which fibre dependence is perturbatively small.

Let

$$\pi : \Sigma \rightarrow \mathbb{CP}^{n-1}$$

be the epistemic projection, and let $V_x := \ker(d\pi_x) \subset T_x\Sigma$ denote the vertical tangent space at $x \in \Sigma$. A Hamiltonian $H : \Sigma \rightarrow \mathbb{R}$ is called *quantum-effective with accuracy ϵ* on a region $\mathcal{S} \subset \Sigma$ if there exists a smooth function

$$h : \mathbb{CP}^{n-1} \rightarrow \mathbb{R}$$

such that, on $\mathcal{S}$,

$$H = h \circ \pi + \delta H,$$

with

$$\sup_{x \in \mathcal{S}} \|d(\delta H)_x|_{V_x}\| \leq \epsilon.$$

Equivalently, the variation of H along the fibres of π is bounded by $O(\epsilon)$ throughout the sector $\mathcal{S}$.

In the limit $\epsilon = 0$, this reduces to exact fibre-invariance,

$$H = h \circ \pi,$$

and the pushforward vector field π_*X_H is exactly well defined on $\mathbb{CP}^{n-1}$. For $\epsilon > 0$ but small, the projected dynamics closes only approximately, with errors controlled by ϵ over the operational timescales of the quantum-effective sector.

Accordingly, the reconstruction in this paper applies to those regimes in which fibre-sensitive corrections remain perturbatively small after projection and coarse-graining. Hamiltonians outside this sector need not induce holomorphic flows on $\mathbb{CP}^{n-1}$ and therefore need not reproduce standard quantum dynamics at the epistemic level.

A.6 Axiom A6: Composite Systems and Marginal Stability

For systems A and B , the joint ontic manifold Σ_{AB} is not assumed to factorise:

$$\Sigma_{AB} \neq \Sigma_A \times \Sigma_B.$$

There exists a projection

$$\pi_{AB} : \Sigma_{AB} \longrightarrow \mathbb{CP}^{n_A-1} \times \mathbb{CP}^{n_B-1},$$

with marginals

$$\pi_A = \text{pr}_A \circ \pi_{AB}, \quad \pi_B = \text{pr}_B \circ \pi_{AB}.$$

The marginals satisfy marginal stability: the epistemic state of each subsystem is independent of measurement choices on the other subsystem. Reduced epistemic densities are obtained through fibre integration:

$$\rho_A = (\pi_A)_* \rho_{AB},$$

which reproduces the mixed-state structure associated with finite-dimensional density operators. Additional details and examples of subsystem marginalisation appear in Appendix F and in Paper D.

A.7 Axiom A7: Measurement Partitions

A measurement interaction with context M induces a family of epistemic outcome regions

$$\{\Omega_i(M)\}_{i \in I} \subset \mathbb{CP}^{n-1},$$

where each $\Omega_i(M)$ is Borel measurable, the family partitions $\mathbb{CP}^{n-1}$ up to μ_{FS} -null sets, and each boundary satisfies

$$\mu_{FS}(\partial\Omega_i(M)) = 0.$$

Thus Axiom A7 requires context-dependent measurable outcome regions with μ_{FS} -null boundaries. No specific geometric construction is assumed at the level of the axiom.

The outcome weight associated with outcome i is defined by

$$P(i \mid M, \rho^{\text{ep}}) = \int_{\Omega_i(M)} \rho^{\text{ep}} d\mu_{FS}.$$

For sharply prepared pure states, the canonical idealisation used in this paper is the rank-one density operator

$$\hat{\rho}_{[\psi]} = |\psi\rangle\langle\psi|.$$

Probabilities are then computed as

$$P(i \mid \hat{\rho}_{[\psi]}, M) = \text{Tr}(\hat{\rho}_{[\psi]} \Pi_i(M)).$$

When an epistemic density representation is used, the equivalent measure-theoretic form is

$$P(i \mid M, \rho^{\text{ep}}) = \int_{\Omega_i(M)} \rho^{\text{ep}} d\mu_{FS}.$$

No localisation-limit formula on shrinking neighbourhoods in $\mathbb{CP}^{n-1}$ is used in the present paper.

For finite-dimensional projective measurements, Fubini–Study Voronoi constructions provide a canonical exemplar of outcome regions satisfying the A7 requirements. Their regularity and null-boundary properties are analysed in Appendix G. However, no claim is made here that all physically realised measurement contexts must induce Voronoi partitions.

Sequential measurement structure, conditionalisation, and the role of context dependence are developed in Appendix G. These ingredients supply the abstract geometric and measure-theoretic foundation required for the finite-dimensional reconstruction presented in the main text.

B Stern–Gerlach Measurement for a Spin 1/2

This appendix presents a Stern–Gerlach (SG) measurement model for a spin- $\frac{1}{2}$ system within the CSD framework. The construction uses classical apparatus fields and Hamiltonian flow on the ontic manifold, then defines outcome regions on $\mathbb{CP}^1$ via Fubini–Study geometry, in accordance with Axiom A7 and Appendix G. Operator language is introduced only as an interpretive shorthand after the geometric structure is established.

B.1 Classical Stern–Gerlach Apparatus

Consider a neutral particle with magnetic moment vector $\boldsymbol{\mu}$ and mass m entering an inhomogeneous magnetic field $\mathbf{B}(\mathbf{r})$ with strong gradient along the z axis,

$$\mathbf{B}(\mathbf{r}) \approx B_0 \hat{\mathbf{z}} + Gz \hat{\mathbf{z}},$$

where $G = \partial B_z / \partial z$ is approximately constant across the beam profile. The classical interaction energy is

$$H_{\text{int}}(\mathbf{r}, \boldsymbol{\mu}) = -\boldsymbol{\mu} \cdot \mathbf{B}(\mathbf{r}),$$

and the resulting force is

$$\mathbf{F}(\mathbf{r}, \boldsymbol{\mu}) = \nabla(\boldsymbol{\mu} \cdot \mathbf{B}(\mathbf{r})) \approx \mu_z G \hat{\mathbf{z}}.$$

For a finite interaction time Δt in the field region, this produces a momentum kick $\Delta p_z \approx \mu_z G \Delta t$ and a subsequent spatial separation of trajectories along z at the detection screen.

B.2 Ontic Manifold for the SG Setup

The ontic manifold for the combined system (internal magnetic degree of freedom plus transverse motion) is taken to be

$$\Sigma_{\text{SG}} = \mathbb{CP}^1 \times T^*\mathbb{R}^3,$$

equipped with the product symplectic form. The $\mathbb{CP}^1$ factor represents the internal epistemic state of the magnetic moment, consistent with the general $\mathbb{CP}^{n-1}$ construction in Appendix C, while $T^*\mathbb{R}^3$ represents position and momentum. The full Hamiltonian is

$$H = H_{\text{free}} + H_{\text{int}},$$

where H_{free} describes free propagation and H_{int} is the classical interaction from Section B.1, lifted to Σ_{SG} .

Under the Hamiltonian flow Φ_t , an initially narrow packet in $T^*\mathbb{R}^3$ splits into two well separated spatial beams corresponding to the sign of the effective longitudinal magnetic moment. This separation is purely classical and defines two disjoint regions in position space at the screen, which we label as the upper and lower macroscopic outcomes.

B.3 Asymptotic Outcome Rays on $\mathbb{CP}^1$

The coupling between $\boldsymbol{\mu}$ and $\mathbf{B}(\mathbf{r})$ also separates ontic trajectories in the internal configuration. In the long-time limit after the magnet, the ontic trajectories that reach the upper spot form a narrow bundle in $\mathbb{CP}^1$ concentrated in a neighbourhood of a ray $[\psi_\uparrow]$, while those reaching the lower spot concentrate near a ray $[\psi_\downarrow]$. These rays are defined operationally:

1. $[\psi_\uparrow]$ is the epistemic state whose ontic fibre, when propagated through the SG apparatus, yields trajectories that almost always arrive in the upper detection region.
2. $[\psi_\downarrow]$ is the epistemic state defined analogously for the lower detection region.

In the standard operator representation these rays correspond to the two eigenstates of the spin component aligned with the apparatus axis, but this identification is derived from the dynamics rather than assumed.

B.4 Fubini–Study Voronoi Outcome Regions on $\mathbb{CP}^1$

Given the two asymptotic outcome rays $[\psi_\uparrow]$ and $[\psi_\downarrow]$, Axiom A7 specifies that the epistemic outcome regions $\Omega_\uparrow$ and $\Omega_\downarrow$ on $\mathbb{CP}^1$ are defined by Fubini–Study Voronoi geometry:

$$\Omega_\uparrow = \{[\psi] \in \mathbb{CP}^1 : d_{\text{FS}}([\psi], [\psi_\uparrow]) < d_{\text{FS}}([\psi], [\psi_\downarrow])\},$$

$$\Omega_\downarrow = \{[\psi] \in \mathbb{CP}^1 : d_{\text{FS}}([\psi], [\psi_\downarrow]) < d_{\text{FS}}([\psi], [\psi_\uparrow])\},$$

with the equality set forming the boundary. Appendix G shows that this boundary has μ_{FS} measure zero. In Bloch-sphere coordinates on $\mathbb{CP}^1$ (Appendix C), with $[\psi_\uparrow]$ and $[\psi_\downarrow]$ at opposite poles, these regions coincide with the two hemispheres separated by the great circle orthogonal to the apparatus axis.

Figure 3: Conceptual depiction of the projection $\pi : \Sigma \rightarrow \mathbb{CP}^{n-1}$ and the associated outcome-region partition on projective state space.

B.5 Born Weights for the SG Outcomes

Let the Stern–Gerlach context determine the two outcome regions $\Omega_{\uparrow}$ and $\Omega_{\downarrow}$ on $\mathbb{CP}^1$ as defined in Section B.4. The probability rule used in this paper is the canonical epistemic measure form

$$P(i \mid M, \rho^{ep}) = \int_{\Omega_i(M)} \rho^{ep} d\mu_{FS}.$$

For a sharply prepared pure state, the canonical idealisation is

$$\hat{\rho}_{[\psi]} = |\psi\rangle\langle\psi|,$$

with probabilities

$$P(i \mid \hat{\rho}_{[\psi]}, M) = \text{Tr}(\hat{\rho}_{[\psi]} \Pi_i(M)).$$

In the spin- $\frac{1}{2}$ Stern–Gerlach setting, the two asymptotic rays $[\psi_{\uparrow}]$ and $[\psi_{\downarrow}]$ correspond, in the standard operator representation, to the two eigenrays of the measured spin component. Hence

$$P(\uparrow \mid \psi) = |\langle\psi_{\uparrow} \mid \psi\rangle|^2 = \cos^2\left(\frac{\theta}{2}\right), \quad P(\downarrow \mid \psi) = |\langle\psi_{\downarrow} \mid \psi\rangle|^2 = \sin^2\left(\frac{\theta}{2}\right),$$

where θ is the Fubini–Study angle between $[\psi]$ and $[\psi_{\uparrow}]$.

Accordingly, the role of the Voronoi construction in the present appendix is to provide an explicit geometric realisation of the measurable outcome regions associated with the Stern–Gerlach apparatus. The Born weights are not identified here with raw intersections of shrinking neighbourhoods in $\mathbb{CP}^1$, but with the canonical state-dependent probability functional used throughout the paper.

C Spin-1 Stern–Gerlach and the Three–Outcome Partition on $\mathbb{CP}^2$

This appendix presents a geometric Stern–Gerlach (SG) model for a spin-1 system within the CSD framework. The classical apparatus induces three macroscopic spatial beams. These correspond to three asymptotic rays in $\mathbb{CP}^2$, which generate a three-cell Fubini–Study Voronoi partition in accordance with Axiom A7 and Appendix G.

C.1 Classical Stern–Gerlach Apparatus for Spin-1

A spin-1 particle with magnetic moment vector $\boldsymbol{\mu}$ enters an inhomogeneous magnetic field with strong gradient along the z axis,

$$\mathbf{B}(\mathbf{r}) \approx B_0 \hat{\mathbf{z}} + Gz \hat{\mathbf{z}}, \quad G = \frac{\partial B_z}{\partial z}.$$

The classical interaction energy is

$$H_{\text{int}}(\mathbf{r}, \boldsymbol{\mu}) = -\boldsymbol{\mu} \cdot \mathbf{B}(\mathbf{r}),$$

producing a force

$$F_z(\mathbf{r}, \boldsymbol{\mu}) \approx \mu_z G.$$

For a spin-1 system the effective longitudinal magnetic moment μ_z takes three distinct classical values, which produce three spatially separated beams on the detection screen. These define three macroscopic outcome regions labelled $(+1)$, (0) , and (-1) .

C.2 Ontic Manifold for the Spin-1 SG Setup

The ontic manifold for the combined internal and spatial degrees of freedom is

$$\Sigma_{\text{SG}}^{(1)} = \mathbb{CP}^2 \times T^*\mathbb{R}^3,$$

with product symplectic structure. The $\mathbb{CP}^2$ factor represents the internal epistemic configuration of the magnetic degree of freedom, consistent with Appendix C. The total Hamiltonian is

$$H = H_{\text{free}} + H_{\text{int}},$$

with H_{free} generating free propagation and H_{int} defined in Section C.1.

Under the Hamiltonian flow Φ_t , initially narrow spatial packets split into three disjoint spatial beams corresponding to the three possible signs and magnitudes of the effective longitudinal moment. These define the three asymptotic macroscopic outcomes used below.

C.3 Asymptotic Outcome Rays in $\mathbb{CP}^2$

The SG interaction couples spatial degrees of freedom to the internal configuration in $\mathbb{CP}^2$. After the particle leaves the magnet, the ontic trajectories associated with the three spatial beams form three narrow bundles in $\mathbb{CP}^2$, concentrated near three rays

$$[\psi_{+1}], \quad [\psi_0], \quad [\psi_{-1}] \in \mathbb{CP}^2.$$

These rays are defined operationally:

1. $[\psi_{+1}]$ is the epistemic ray whose ontic fibre yields trajectories predominantly reaching the upper detection region.
2. $[\psi_0]$ corresponds to the middle beam.
3. $[\psi_{-1}]$ corresponds to the lower beam.

No operator representation is required for their definition. Under the standard quantum representation, these correspond to the three eigenrays of the spin projection along the apparatus axis, but this identification is derived rather than assumed.

C.4 Fubini–Study Voronoi Partition for Three Rays

Given the three rays defined above, Axiom A7 specifies that the outcome regions on $\mathbb{CP}^2$ are given by the Fubini–Study Voronoi partition:

$$\Omega_i = \{[\psi] \in \mathbb{CP}^2 : d_{\text{FS}}([\psi], [\psi_i]) < d_{\text{FS}}([\psi], [\psi_j]) \text{ for all } j \neq i\}, \quad i \in \{+1, 0, -1\}.$$

The boundaries where two or more distances coincide form codimension–1 submanifolds. Appendix G shows these boundaries have μ_{FS} measure zero, ensuring well defined probabilities.

Geometric interpretation. In homogeneous coordinates $[z_1 : z_2 : z_3]$, the three rays occupy the vertices of a projective triangle in $\mathbb{CP}^2$. The outcome regions correspond to the minimal FS–distance cells associated with these vertices. Unlike the $\mathbb{CP}^1$ (Bloch sphere) case, the cells are not hemispheres but curved two–dimensional domains bounded by geodesic arcs determined by equal–distance loci.

C.5 Three–Outcome Born Weights

Let the spin–1 Stern–Gerlach context determine the three measurable outcome regions

$$\Omega_{+1}, \Omega_0, \Omega_{-1} \subset \mathbb{CP}^2$$

as defined in Section C.4. The probability rule used in this paper is

$$P(i \mid M, \rho^{ep}) = \int_{\Omega_i(M)} \rho^{ep} d\mu_{\text{FS}}.$$

For a sharply prepared pure state, the canonical idealisation is

$$\hat{\rho}_{[\psi]} = |\psi\rangle\langle\psi|,$$

with probabilities

$$P(i \mid \hat{\rho}_{[\psi]}, M) = \text{Tr}(\hat{\rho}_{[\psi]} \Pi_i(M)), \quad i \in \{+1, 0, -1\}.$$

In the standard operator representation, the asymptotic rays $[\psi_{+1}], [\psi_0], [\psi_{-1}]$ correspond to the three eigenrays of the spin projection along the apparatus axis. Hence

$$P(i \mid \psi) = |\langle \psi_i \mid \psi \rangle|^2, \quad i \in \{+1, 0, -1\}.$$

The geometric content of the present construction is that the Stern–Gerlach apparatus induces a concrete three-cell measurable partition on $\mathbb{CP}^2$, realised here by a Fubini–Study Voronoi decomposition. This appendix does not identify the Born weights with raw neighbourhood-intersection volumes for sharply localised states; rather, it uses the canonical state-dependent probability rule established in the main reconstruction.

D Fubini–Study Voronoi Geometry as a Projective Exemplar

This appendix studies Fubini–Study Voronoi geometry as a canonical finite-dimensional exemplar of the abstract measurement-partition structure postulated in Axiom A7. The axiom itself requires only context-dependent measurable outcome regions with μ_{FS} -null boundaries on $\mathbb{CP}^{n-1}$. In the projective case, Voronoi constructions provide a natural and explicit family of such regions, making it possible to analyse measurability, boundary structure, and probability assignments in concrete geometric terms. No claim is made here that all admissible or physically realised measurement partitions must take Voronoi form.

D.1 Measurement Contexts and Reference Rays

A measurement with m outcomes is represented by m distinct rays

$$[\psi_1], \dots, [\psi_m] \in \mathbb{CP}^{n-1}.$$

These rays arise operationally from the asymptotic behaviour of a physical apparatus, as illustrated in Appendices B and C. No operator structure is assumed in their definition.

D.2 Fubini–Study Voronoi Cells

In this projective exemplar, the outcome regions are realised by the FS–Voronoi cells:

$$\Omega_i = \left\{ [\psi] \in \mathbb{CP}^{n-1} : d_{FS}([\psi], [\psi_i]) < d_{FS}([\psi], [\psi_j]) \quad \forall j \neq i \right\}.$$

The FS distance is

$$d_{FS}([\phi], [\psi]) = \arccos \left(\frac{|\langle \phi \mid \psi \rangle|}{\|\phi\| \|\psi\|} \right),$$

using any choice of representatives in $\mathbb{C}^n$.

D.3 Boundary Surfaces and Equal–Distance Loci

The boundary between Ω_i and Ω_j is the equal–distance set

$$\partial_{ij} = \left\{ [\psi] : d_{FS}([\psi], [\psi_i]) = d_{FS}([\psi], [\psi_j]) \right\}.$$

This defines a smooth codimension–1 submanifold except on a lower–dimensional singular set where gradients vanish. Appendix G shows that both the smooth and singular parts of ∂_{ij} have μ_{FS} measure zero.

D.4 Geometry of Voronoi Cells

For $m = 2$, the cells correspond to hemispheres in $\mathbb{CP}^1$ (Appendix B). For $m = 3$ in $\mathbb{CP}^2$, they form curved regions bounded by geodesic arcs (Appendix C). For general m and n , each Ω_i is a connected open region defined entirely by FS–distance comparisons.

Equal–distance hypersurfaces in projective space are generally curved rather than totally geodesic. In the projective examples studied here, this geometry is relevant for the analysis of context-dependent partition structure.

D.5 Degenerate Measurements

If the outcome i corresponds to a subspace $\mathcal{H}_i$ of dimension $d_i > 1$, choose any orthonormal basis $\{\psi_{i\ell}\}_{\ell=1}^{d_i}$ of $\mathcal{H}_i$. Construct Voronoi cells for these basis rays and union them:

$$\Omega_i = \bigcup_{\ell=1}^{d_i} \Omega_{i\ell}.$$

Appendix E shows that Ω_i is independent of the basis choice up to null-measure discrepancies.

D.6 Measurability of Outcome Regions

Each Ω_i is open, and its boundary is a finite union of codimension-1 null-measure sets. Therefore Ω_i is Borel-measurable, and for any absolutely continuous epistemic density ρ (Axiom A4) the probability

$$P(i \mid \rho) = \int_{\Omega_i} \rho d\mu_{\text{FS}}$$

is well defined.

E Degenerate Projective Measurements as Nearest-Subspace Partitions

This appendix studies degenerate projective measurements as a further finite-dimensional exemplar of the abstract measurement-partition structure postulated in Axiom A7. The axiom itself requires only context-dependent measurable outcome regions with μ_{FS} -null boundaries. In the degenerate projective case, Fubini-Study geometry provides a natural construction of such regions through nearest-subspace characterisation, with equivalent nearest-ray Voronoi realisations up to μ_{FS} -null sets. We also establish basis independence up to null sets.

E.1 Degenerate Subspaces as Measurement Outcomes

Let a projective measurement have outcomes labelled by mutually orthogonal subspaces

$$\mathcal{H}_i \subset \mathbb{C}^n, \quad \dim \mathcal{H}_i = d_i > 1.$$

The epistemic representation of each outcome is the projective submanifold

$$\mathbb{P}(\mathcal{H}_i) \subset \mathbb{CP}^{n-1}.$$

Choose any orthonormal basis

$$\psi_{i1}, \dots, \psi_{id_i}$$

for $\mathcal{H}_i$, and denote their projective images by $[\psi_{i\ell}]$.

E.2 Voronoi Realisation of the Nearest-Subspace Partition

Choosing an orthonormal basis for each outcome subspace, one may realise the degenerate partition by the union of Voronoi cells associated with the corresponding basis rays, as in Appendix D:

$$\Omega_{i\ell} = \left\{ [\psi] : d_{\text{FS}}([\psi], [\psi_{i\ell}]) < d_{\text{FS}}([\psi], [\psi_{jk}]) \quad \forall (j, k) \neq (i, \ell) \right\}.$$

The outcome region associated with the degenerate subspace $\mathcal{H}_i$ is

$$\Omega_i = \bigcup_{\ell=1}^{d_i} \Omega_{i\ell}.$$

E.3 Basis-Independence of Outcome Regions

Let $\{\psi_{i\ell}\}$ and $\{\phi_{i\ell}\}$ be two orthonormal bases of $\mathcal{H}_i$. Let Ω_i and $\tilde{\Omega}_i$ denote the corresponding outcome regions.

The only points whose assignment may change lie on equal-distance boundaries of the form

$$d_{\text{FS}}([\psi], [\psi_{i\ell}]) = d_{\text{FS}}([\psi], [\psi_{im}]).$$

Appendix G shows that such surfaces have μ_{FS} measure zero. Therefore

$$\mu_{\text{FS}}(\Omega_i \Delta \tilde{\Omega}_i) = 0,$$

so the outcome regions are basis independent up to null-measure differences.

E.4 Nearest-Subspace Characterisation

Let Π_i denote the orthogonal projector onto $\mathcal{H}_i$. The same outcome region can be described equivalently in nearest-subspace form as

$$\Omega_i = \left\{ [\psi] : \|\Pi_i \psi\| > \|\Pi_j \psi\| \quad \forall j \neq i \right\}.$$

This follows from the fact that the FS distance from a point to a subspace is minimised by the ray in the subspace with maximal projection amplitude. Thus the nearest-subspace condition and nearest-ray Voronoi construction yield identical regions up to null sets.

E.5 Measurability

Each Ω_i is open, because the inequalities $\|\Pi_i \psi\| > \|\Pi_j \psi\|$ define open sets. The boundary set

$$\partial\Omega_i = \bigcup_{j \neq i} \left\{ [\psi] : \|\Pi_i \psi\| = \|\Pi_j \psi\| \right\},$$

is a finite union of codimension-1 FS-null sets (Appendix G). Therefore Ω_i is Borel measurable, and the probability assignment

$$P(i \mid \rho) = \int_{\Omega_i} \rho \, d\mu_{\text{FS}}$$

is well defined for any epistemic density satisfying Axiom A4.

F Sequential Measurements and Context Transitions

This appendix describes the geometric structure of sequential measurements in the CSD framework. A sequential measurement involves two measurement contexts M and N , each determining outcome regions on CP^{n-1} in the sense of Axiom A7. In the finite-dimensional projective examples studied here, these regions are realised explicitly by Fubini-Study Voronoi constructions and their degenerate nearest-subspace extensions, as developed in Appendices D and E. The goal is to formalise (i) conditioning after the first outcome and (ii) how incompatibility of the induced partitions leads to order dependence.

F.1 Measurement Contexts

Let context M be specified by rays

$$[\psi_i(M)] \in CP^{n-1}, \quad i = 1, \dots, m,$$

together with the associated outcome regions $\Omega_i(M)$. In the projective examples studied here, these regions are realised by FS-distance minimisation relative to the chosen rays. A second context N is defined analogously, with rays $[\psi_j(N)]$ and outcome regions $\Omega_j(N)$.

F.2 First Measurement: Conditioning

Given an initial epistemic density ρ absolutely continuous with respect to μ_{FS} , the conditional density after observing outcome i of M is

$$\rho_i^M([\psi]) = \frac{\rho([\psi]) \chi_{\Omega_i(M)}([\psi])}{\int_{\Omega_i(M)} \rho \, d\mu_{\text{FS}}}.$$

No assumption is made that ρ_i^M is uniform on $\Omega_i(M)$; its form depends on the initial ρ and the geometry of the region.

F.3 Second Measurement: Transition Probabilities

The probability of obtaining outcome j of context N after observing i of M is

$$P(j \mid i; N, M) = \int_{\Omega_j(N)} \rho_i^M([\psi]) \, d\mu_{\text{FS}}.$$

This integral is well defined because the sets $\Omega_j(N)$ are Borel measurable (Appendix G).

F.4 Geometric Incompatibility of Contexts

Contexts M and N are *compatible* if each $\Omega_i(M)$ is, up to null sets, contained in a union of outcome regions for N :

$$\mu_{\text{FS}} \left(\Omega_i(M) \setminus \bigcup_{j \in J(i)} \Omega_j(N) \right) = 0.$$

If this does not hold, the contexts are *geometrically incompatible*. Incompatible contexts have nontrivial overlaps

$$\mu_{\text{FS}}(\Omega_i(M) \cap \Omega_j(N)) \neq 0,$$

which generically implies order dependence.

F.5 Sequential Geometry on CP^1 and CP^2

For CP^1 (spin- $\frac{1}{2}$), contexts correspond to pairs of hemispheres on the Bloch sphere. Two contexts are compatible if and only if their axes coincide. Otherwise their hemispherical partitions slice through each other, producing incompatible overlaps.

For CP^2 (spin-1), each context corresponds to three Voronoi cells. Generic choices of rays yield partitions whose boundaries curve through those of the other context, making compatibility rare.

F.6 Contextuality as Partition Nonrefinement

CSD does not assume noncontextual hidden variables. Contextuality arises from the incompatibility of context-dependent outcome partitions:

$$M \text{ and } N \text{ incompatible} \implies \{\Omega_i(M)\} \text{ and } \{\Omega_j(N)\} \text{ admit no common refinement (up to null sets).}$$

This expresses *operational contextuality*. For $n = 2$, this leads to order dependence but not Kochen–Specker (KS) contradictions. For $n \geq 3$, geometric incompatibility is necessary but not sufficient for KS-type no-go theorems.

G Measure Theory and Measurability on CP^{n-1}

This appendix establishes the measure-theoretic foundations used throughout Paper C. The Fubini–Study (FS) metric induces a canonical finite Borel measure μ_{FS} on CP^{n-1} , and all epistemic densities are assumed to be absolutely continuous with respect to this measure (Axiom A4). We prove regularity, null-measure properties of Voronoi boundaries, and measurability of all sets used in probability assignments.

G.1 Fubini–Study Measure on CP^{n-1}

The FS metric on CP^{n-1} is the unique $\text{SU}(n)$ -invariant Kähler metric. Its volume form $\omega^{n-1}/(n-1)!$ defines a finite Borel measure

$$\mu_{\text{FS}}(A) = \int_A \frac{\omega^{n-1}}{(n-1)!}, \quad A \subset \text{CP}^{n-1}.$$

Because CP^{n-1} is compact, μ_{FS} is finite. It is regular and invariant under the natural $\text{SU}(n)$ action.

G.2 Borel Structure

Let $\mathcal{B}$ be the Borel σ -algebra generated by FS-open sets. Since CP^{n-1} is compact and second-countable:

- open and closed sets are measurable,
- boundaries of open sets are measurable,
- countable unions and intersections of Borel sets are measurable,
- all compact sets have finite measure.

All outcome regions introduced in Appendices B–G belong to $\mathcal{B}$.

G.3 Equal-Distance Surfaces and Null Sets

For two rays $[\psi_i]$ and $[\psi_j]$, the equal-distance surface is

$$\partial_{ij} = \left\{ [\phi] : d_{\text{FS}}([\phi], [\psi_i]) = d_{\text{FS}}([\phi], [\psi_j]) \right\}.$$

Let

$$F(\phi) = \cos d_{\text{FS}}([\phi], [\psi_i]) - \cos d_{\text{FS}}([\phi], [\psi_j]).$$

Where $\nabla F \neq 0$, the implicit function theorem shows ∂_{ij} is a smooth codimension-1 submanifold. Its FS measure is zero.

The singular set $\{\nabla F = 0\}$ has codimension at least 2, so it is also μ_{FS} -null. Therefore

$$\mu_{\text{FS}}(\partial_{ij}) = 0.$$

Finite unions of such surfaces are also null.

G.4 Voronoi Cell Boundaries

For a measurement context, the boundary of each outcome region satisfies

$$\partial\Omega_i \subseteq \bigcup_{j \neq i} \partial_{ij}.$$

Since each ∂_{ij} is μ_{FS} -null,

$$\mu_{\text{FS}}(\partial\Omega_i) = 0.$$

Thus outcome boundaries do not contribute to probabilities.

G.5 Absolute Continuity of Epistemic Densities

Axiom A4 requires ρ to satisfy

$$\rho \in L^1(\mathbb{CP}^{n-1}, \mu_{\text{FS}}), \quad \rho \geq 0, \quad \int \rho d\mu_{\text{FS}} = 1.$$

Absolute continuity ensures

$$\mu_{\text{FS}}(A) = 0 \implies \int_A \rho d\mu_{\text{FS}} = 0,$$

so boundary sets never affect probabilities.

G.6 Invariance Under Null Modifications

The null-boundary condition implies that outcome probabilities do not depend on how boundary points are assigned, provided the epistemic measure is absolutely continuous with respect to μ_{FS} .

Lemma G.6: Let ν^{ep} be a probability measure on $\mathbb{CP}^{n-1}$ such that

$$\nu^{\text{ep}} \ll \mu_{\text{FS}}.$$

Let $\Omega_i, \tilde{\Omega}_i \subset \mathbb{CP}^{n-1}$ be Borel sets satisfying

$$\mu_{\text{FS}}(\Omega_i \Delta \tilde{\Omega}_i) = 0,$$

where Δ denotes symmetric difference. Then

$$\nu^{\text{ep}}(\Omega_i) = \nu^{\text{ep}}(\tilde{\Omega}_i).$$

Proof: Since

$$\Omega_i \Delta \tilde{\Omega}_i = (\Omega_i \setminus \tilde{\Omega}_i) \cup (\tilde{\Omega}_i \setminus \Omega_i),$$

the hypothesis implies that both difference sets are μ_{FS} -null. Because $\nu^{\text{ep}} \ll \mu_{FS}$, they are also ν^{ep} -null. Hence

$$\nu^{\text{ep}}(\Omega_i \setminus \tilde{\Omega}_i) = 0, \quad \nu^{\text{ep}}(\tilde{\Omega}_i \setminus \Omega_i) = 0,$$

so

$$\nu^{\text{ep}}(\Omega_i) = \nu^{\text{ep}}(\tilde{\Omega}_i).$$

□

In particular, if the only differences between two outcome-region assignments lie on μ_{FS} -null boundaries, then they define identical probabilities for all absolutely continuous epistemic measures.

G.7 Conditional Densities and Sequential Measurements

For any Borel set A with $\mu_{FS}(A) > 0$,

$$\rho_A([\psi]) = \frac{\rho([\psi]) \chi_A([\psi])}{\int_A \rho \, d\mu_{FS}}$$

defines a valid conditional density.

Boundaries are μ_{FS} -null, so conditioning is unaffected by the choice of representative of A . Sequential measurement probabilities in Appendix G follow directly:

$$P(j \mid i; N, M) = \int_{\Omega_j(N)} \rho_{\Omega_i(M)} \, d\mu_{FS}.$$

G.8 Stability Under Regular Context Variation

We now record a basic robustness property of the probability assignment under sufficiently regular variation of measurement context.

Proposition G.8: Let $\lambda \mapsto M(\lambda)$ be a one-parameter family of measurement contexts, and for each λ let

$$\{\Omega_i(M(\lambda))\}_{i \in I}$$

be a family of Borel outcome regions on $\mathbb{CP}^{n-1}$. Assume:

1. for each i and λ , the boundary satisfies

$$\mu_{FS}(\partial\Omega_i(M(\lambda))) = 0;$$

2. for each fixed i , the indicator functions

$$\chi_{\Omega_i(M(\lambda))}$$

converge pointwise μ_{FS} -almost everywhere as $\lambda \rightarrow \lambda_0$;

3. $\nu^{\text{ep}} \ll \mu_{FS}$.

Then

$$\nu^{\text{ep}}(\Omega_i(M(\lambda))) \longrightarrow \nu^{\text{ep}}(\Omega_i(M(\lambda_0))) \quad \text{as } \lambda \rightarrow \lambda_0.$$

Proof: By assumption,

$$\chi_{\Omega_i(M(\lambda))}([\psi]) \rightarrow \chi_{\Omega_i(M(\lambda_0))}([\psi])$$

for μ_{FS} -almost every $[\psi]$. Since indicator functions are bounded by 1, the dominated convergence theorem gives

$$\int_{\mathbb{CP}^{n-1}} \chi_{\Omega_i(M(\lambda))} \, d\nu^{\text{ep}} \rightarrow \int_{\mathbb{CP}^{n-1}} \chi_{\Omega_i(M(\lambda_0))} \, d\nu^{\text{ep}}.$$

But these integrals are exactly

$$\nu^{\text{ep}}(\Omega_i(M(\lambda))) \quad \text{and} \quad \nu^{\text{ep}}(\Omega_i(M(\lambda_0))),$$

respectively. This proves the claim. □

This proposition formalises the fact that outcome probabilities are stable under sufficiently regular deformations of context. In the finite-dimensional projective examples studied here, such regularity is supplied by smooth variation of the defining rays together with the null-boundary results of Sections G.3 and G.4.

H Schrödinger Dynamics as Effective Projected Hamiltonian Flow

This appendix explains how standard finite-dimensional Schrödinger dynamics arises as the leading epistemic evolution induced by deterministic Hamiltonian flow on the ontic manifold Σ . In the present version of the reconstruction, this emergence is not tied to an exact global fibre-invariance condition. Instead, it holds within the *quantum-effective sector* introduced in Axiom A5 and formalised in Appendix I, where fibre-sensitive corrections remain perturbatively small over the operational timescale of interest.

H.1 Hamiltonian flow on Σ

Let (Σ, ω) be the compact Kähler manifold of Appendix A, and let

$$H : \Sigma \rightarrow \mathbb{R}$$

be a smooth ontic Hamiltonian. Its Hamiltonian vector field X_H is defined by

$$\iota_{X_H} \omega = dH,$$

and generates a deterministic flow Φ_t on Σ ,

$$\frac{dx}{dt} = X_H(x).$$

Because Hamiltonian flow preserves the symplectic volume form, the associated Liouville measure μ_L is invariant under Φ_t .

The epistemic description is obtained through the projection

$$\pi : \Sigma \rightarrow \mathbb{CP}^{n-1},$$

which identifies many ontic microstates with the same epistemic ray. It follows that a projected dynamics on $\mathbb{CP}^{n-1}$ does not exist automatically for arbitrary H . Additional control on the fibre dependence of H is required.

H.2 Exact and approximate projectability

For $[\psi] = \pi(x)$, let

$$F_{[\psi]} := \pi^{-1}([\psi])$$

denote the fibre above the epistemic point $[\psi]$. In the exactly projectable case, one has

$$H = h \circ \pi$$

for some smooth function

$$h : \mathbb{CP}^{n-1} \rightarrow \mathbb{R}.$$

Then H is constant along each fibre, and the projected vector field is exactly well defined.

In the present paper, however, the relevant assumption is weaker. Within the quantum-effective sector, we allow

$$H = h \circ \pi + \delta H,$$

where the fibre-sensitive correction δH is small in the sense made precise in Appendix I. Writing

$$V_x := \ker(d\pi_x) \subset T_x \Sigma,$$

the effective projectability condition is that

$$\|d(\delta H)_x|_{V_x}\| = O(\epsilon), \quad \epsilon \ll 1,$$

throughout the dynamically relevant region and over the operational time interval of interest.

Thus exact fibre-invariance is recovered only in the ideal limit $\epsilon = 0$. More generally, the projected epistemic evolution is defined only up to controlled corrections of order $O(\epsilon)$.

H.3 Projected Hamiltonian dynamics on $\mathbb{CP}^{n-1}$

Let Y_h denote the Hamiltonian vector field on $\mathbb{CP}^{n-1}$ generated by h with respect to the Fubini–Study symplectic form ω_{FS} :

$$\iota_{Y_h} \omega_{FS} = dh.$$

In the exactly projectable case, the pushforward of the ontic Hamiltonian flow coincides with Y_h , so that the induced dynamics on $\mathbb{CP}^{n-1}$ is a genuine Hamiltonian flow preserving the Fubini–Study geometry.

More generally, Appendix I shows that if H lies in the quantum-effective sector, then the projected trajectory

$$z_t := \pi(x_t), \quad x_t = \Phi_t(x_0),$$

satisfies

$$\dot{z}_t = Y_h(z_t) + r_t, \quad \|r_t\| = O(\epsilon),$$

and hence remains within $O(\epsilon)$ of the Hamiltonian flow generated by h on $\mathbb{CP}^{n-1}$ for finite operational times. In this sense, the epistemic dynamics is Hamiltonian to leading order, with controlled deviations determined by the small fibre-sensitive part of the ontic Hamiltonian.

Accordingly, the geometric structures used in finite-dimensional quantum theory, namely the symplectic form, the compatible complex structure, and the Fubini–Study measure, are preserved exactly only in the limit $\epsilon = 0$, but remain valid as an effective description whenever $\epsilon \ll 1$.

H.4 Quadratic Hamiltonians on $\mathbb{CP}^{n-1}$

A smooth function h on $\mathbb{CP}^{n-1}$ is called *quadratic* if it is induced by a Hermitian operator $\hat{H}$ on $\mathbb{C}^n$ through

$$h([\psi]) = \frac{\langle \psi, \hat{H} \psi \rangle}{\langle \psi, \psi \rangle}.$$

Such functions generate the standard unitary Hamiltonian flows on projective Hilbert space (Ashtekar and Schilling 1999; Bengtsson and Życzkowski 2006).

Equivalently, the Hamiltonian flow of h on $\mathbb{CP}^{n-1}$ is the projective image of the linear Schrödinger equation

$$i\hbar \frac{d}{dt} |\psi\rangle = \hat{H} |\psi\rangle.$$

Thus quadratic Hamiltonians are precisely those whose projected Hamiltonian dynamics reproduces ordinary finite-dimensional quantum evolution.

This point is important conceptually. The Hilbert-space operator $\hat{H}$ is not assumed to be ontically fundamental on Σ . Rather, it appears as the effective generator of the leading projected flow on $\mathbb{CP}^{n-1}$ once the ontic dynamics lies in the quantum-effective sector and the induced epistemic Hamiltonian h is quadratic.

H.5 Emergence of Schrödinger evolution

Suppose now that the ontic Hamiltonian satisfies

$$H = h \circ \pi + \delta H$$

with

$$\|d(\delta H)|_V\| = O(\epsilon), \quad \epsilon \ll 1,$$

throughout the relevant sector, and suppose further that the effective projected Hamiltonian h is quadratic:

$$h([\psi]) = \frac{\langle \psi, \hat{H} \psi \rangle}{\langle \psi, \psi \rangle}.$$

Then, by Appendix I, the projected ontic trajectory obeys

$$\pi(\Phi_t(x)) = \exp\left(-\frac{i}{\hbar} \hat{H} t\right) \pi(x) + O(\epsilon)$$

for finite operational times, where the $O(\epsilon)$ term denotes the controlled deviation from exact projectability.

Hence Schrödinger evolution is recovered as the *leading effective epistemic dynamics* associated with deterministic ontic Hamiltonian flow in the quantum-effective sector. Exact unitary evolution appears in the ideal limit $\epsilon \rightarrow 0$, while laboratory quantum dynamics corresponds to the regime in which the fibre-sensitive corrections remain negligible at the resolution probed by the experiment.

In this framework, unitarity is therefore not a primitive axiom imposed at the ontic level. It is the effective projected expression of deterministic symplectic flow once the ontic Hamiltonian is approximately projectable and the induced epistemic Hamiltonian is quadratic.

H.6 Outside the quantum-effective sector

If H lies outside the quantum-effective sector, then the projected dynamics on $\mathbb{CP}^{n-1}$ need not be well defined even approximately. This can occur if fibre dependence is strong, if fibre-base coupling accumulates over long times, or if the experimental protocol becomes sensitive to ontic structure beyond the projected epistemic coordinates.

Even when an effective projection exists, a general smooth function h on $\mathbb{CP}^{n-1}$ need not be quadratic. Its Hamiltonian flow is then generally not the projectivisation of a linear Schrödinger equation. Such flows correspond to more general geometric dynamics on projective space and are excluded from the present finite-dimensional reconstruction, whose aim is to recover the standard quantum sector.

Accordingly, this paper restricts attention to the regime in which two conditions hold:

1. the ontic Hamiltonian is approximately projectable, with fibre-sensitive corrections bounded by $O(\epsilon)$, and
2. the resulting effective projected Hamiltonian h is quadratic.

Under these conditions, deterministic flow on Σ reproduces standard finite-dimensional quantum dynamics on $\mathbb{CP}^{n-1}$ to leading order. The analysis of more general nonquadratic projected flows, and of mechanisms by which generic ontic Hamiltonians may reduce to the quadratic sector, is deferred to later work.

I Approximate Projectability in the Quantum-Effective Sector

This appendix reformulates the exact fibre-invariance condition used in the main text as a controlled effective approximation. The purpose is to make precise the statement that standard finite-dimensional quantum dynamics is recovered whenever fibre-sensitive corrections remain perturbatively small over the operational timescale of interest.

I.1 Geometric setting

Let

$$\pi : \Sigma \rightarrow \mathbb{CP}^{n-1}$$

be the smooth epistemic projection introduced in Axiom A4. For each $x \in \Sigma$, let

$$V_x := \ker(d\pi_x) \subset T_x \Sigma$$

denote the vertical tangent space, and let $H_x \subset T_x \Sigma$ denote a chosen horizontal complement, so that

$$T_x \Sigma = H_x \oplus V_x.$$

Let X_H be the Hamiltonian vector field generated by a smooth Hamiltonian $H : \Sigma \rightarrow \mathbb{R}$.

In the exactly projectable case one has

$$H = h \circ \pi$$

for some smooth $h : \mathbb{CP}^{n-1} \rightarrow \mathbb{R}$, and the pushforward $d\pi(X_H)$ depends only on the epistemic point $[\psi] = \pi(x)$. The projected dynamics is then well defined on $\mathbb{CP}^{n-1}$.

We now weaken this condition.

I.2 Definition of approximate projectability

Let $\mathcal{S} \subset \Sigma$ be a dynamically relevant region. We say that H is *approximately projectable with accuracy ϵ on $\mathcal{S}$* if there exists a smooth function

$$h : \mathbb{CP}^{n-1} \rightarrow \mathbb{R}$$

such that

$$H = h \circ \pi + \delta H$$

on $\mathcal{S}$, with

$$\sup_{x \in \mathcal{S}} \|d(\delta H)_x|_{V_x}\| \leq \epsilon.$$

Equivalently, the variation of H along the fibres of π is uniformly bounded by $O(\epsilon)$ throughout $\mathcal{S}$.

Write

$$X_H = X_{h \circ \pi} + X_{\delta H}.$$

The first term is exactly projectable. The second term encodes the fibre-sensitive correction.

I.3 Approximate projectability lemma

Lemma I.1 (Approximate projected dynamics). *Let $\mathcal{S} \subset \Sigma$ be compact, and suppose H is approximately projectable with accuracy ϵ on $\mathcal{S}$. Assume further that there exists a constant $C_{\pi, \omega} > 0$, depending only on the local geometry of π and ω on $\mathcal{S}$, such that*

$$\|d\pi_x(X_{\delta H}(x))\| \leq C_{\pi, \omega} \|d(\delta H)_x|_{V_x}\|$$

for all $x \in \mathcal{S}$. Then there exists a smooth vector field Y_h on $\pi(\mathcal{S}) \subset \mathbb{CP}^{n-1}$, namely the Hamiltonian vector field generated by h , such that

$$\|d\pi_x(X_H(x)) - Y_h(\pi(x))\| \leq C\epsilon$$

for all $x \in \mathcal{S}$, where $C := C_{\pi, \omega}$.

Consequently, if x_t is an integral curve of X_H contained in $\mathcal{S}$ for $t \in [0, T]$, and if y_t is the integral curve of Y_h on $\mathbb{CP}^{n-1}$ with initial condition

$$y_0 = \pi(x_0),$$

then

$$d_{\text{FS}}(\pi(x_t), y_t) \leq K_T \epsilon$$

for all $t \in [0, T]$, where $K_T > 0$ depends on T and on a Lipschitz bound for Y_h on $\pi(\mathcal{S})$, but is independent of ϵ .

Proof. Since

$$X_H = X_{h \circ \pi} + X_{\delta H},$$

we obtain

$$d\pi_x(X_H(x)) = d\pi_x(X_{h \circ \pi}(x)) + d\pi_x(X_{\delta H}(x)).$$

By exact projectability of $h \circ \pi$, the first term depends only on $\pi(x)$, and equals the Hamiltonian vector field $Y_h(\pi(x))$ induced on $\mathbb{CP}^{n-1}$. Hence

$$d\pi_x(X_H(x)) - Y_h(\pi(x)) = d\pi_x(X_{\delta H}(x)).$$

The assumed bound gives

$$\|d\pi_x(X_H(x)) - Y_h(\pi(x))\| \leq C_{\pi, \omega} \|d(\delta H)_x|_{V_x}\| \leq C_{\pi, \omega} \epsilon.$$

This proves the first estimate.

For the trajectory estimate, define

$$z_t := \pi(x_t).$$

Then z_t satisfies the perturbed equation

$$\dot{z}_t = Y_h(z_t) + r_t, \quad \|r_t\| \leq C\epsilon.$$

Standard stability estimates for ordinary differential equations on compact manifolds then imply

$$d_{\text{FS}}(z_t, y_t) \leq K_T \epsilon$$

for $t \in [0, T]$, where K_T grows at most exponentially with T through the Lipschitz constant of Y_h . This yields the claimed result. $\square$

I.4 Interpretation

Lemma I.1 shows that exact fibre-invariance is not required in order to recover an effective epistemic evolution. It is sufficient that fibre-dependent corrections remain uniformly small in the relevant sector. In that case the ontic flow on Σ projects to a well-defined Hamiltonian flow on $\mathbb{CP}^{n-1}$ up to errors of order $O(\epsilon)$ over finite operational times.

This is the sense in which the finite-dimensional quantum reconstruction should be understood in the present paper. The emergent Schrödinger dynamics is exact only in the ideal limit $\epsilon = 0$, but remains a valid effective description whenever $\epsilon \ll 1$ on the timescale probed by the experiment.

I.5 Quantum-effective sector

We define the *quantum-effective sector* to be any regime $\mathcal{S} \subset \Sigma$ in which all of the following hold:

1. the Hamiltonian admits a decomposition

$$H = h \circ \pi + \delta H$$

with fibre variation bounded by

$$\sup_{x \in \mathcal{S}} \|d(\delta H)_x|_{V_x}\| \leq \epsilon,$$

for some $\epsilon \ll 1$;

2. the ontic trajectory remains in $\mathcal{S}$ throughout the operational time interval $t \in [0, T]$;
3. the accumulated projection error satisfies

$$K_T \epsilon \ll 1,$$

so that the induced epistemic trajectory remains close to the Hamiltonian flow generated by h on $\mathbb{CP}^{n-1}$;

4. experimental observables depend only on the projected epistemic state to the resolution relevant for the experiment.

Operationally, this sector corresponds to a regime in which fibre-sensitive corrections are either weakly coupled, rapidly averaged, or coarse-grained below experimental resolution. In such a regime the standard finite-dimensional quantum description on $\mathbb{CP}^{n-1}$ is recovered as an effective theory.

I.6 Domain of validity

The approximation above need not hold universally. It may fail in regimes where fibre-base coupling becomes strong, where long-time accumulation invalidates the bound $K_T \epsilon \ll 1$, or where the measurement protocol becomes sensitive to ontic structure beyond the projected epistemic coordinates. For this reason Axiom A5 should be read as a controlled sectoral assumption rather than as a claim about all Hamiltonians on Σ .

References

- Vladimir I. Arnold. *Mathematical Methods of Classical Mechanics*. Springer, New York, 2nd edition, 1989. ISBN 9780387968902.
- Vladimir Igorevich Arnold and A. (André) Avez. *Ergodic problems of classical mechanics*. The mathematical physics monograph series. Benjamin, 1968. URL <https://cir.nii.ac.jp/crid/1130282273312514176>.
- Abhay Ashtekar and Thomas A. Schilling. Geometry of quantum mechanics. In Alex Harvey, editor, *On Einstein's Path: Essays in Honor of Engelbert Schucking*, pages 23–65. Springer, New York, 1997. doi:10.1007/978-1-4612-4058-6_2.
- J. S. Bell. On the einstein podolsky rosen paradox. *Physics Physique Fizika*, 1(3):195–200, November 1964. ISSN 0554-128X. doi:10.1103/physicsphysiquefizika.1.195. URL <http://dx.doi.org/10.1103/PhysicsPhysiqueFizika.1.195>.
- Ingemar Bengtsson and Karol Życzkowski. *Geometry of Quantum States: An Introduction to Quantum Entanglement*. Cambridge University Press, Cambridge, 2006. ISBN 9780521814515.
- Zayn Blore. Volume-based probability: Outcome frequencies from deterministic geometry, 2025a. URL <https://doi.org/10.5281/zenodo.16943198>.
- Zayn Blore. Fixing the measure: Deriving the born rule from symmetry in deterministic geometry, 2025b. URL <https://doi.org/10.5281/zenodo.18094980>.

- David Bohm. A suggested interpretation of the quantum theory in terms of “hidden” variables. i. *Physical Review*, 85(2):166–179, January 1952a. ISSN 0031-899X. doi:10.1103/physrev.85.166. URL <http://dx.doi.org/10.1103/PhysRev.85.166>.
- David Bohm. A suggested interpretation of the quantum theory in terms of “hidden” variables. ii. *Physical Review*, 85(2):180–193, January 1952b. ISSN 0031-899X. doi:10.1103/physrev.85.180. URL <http://dx.doi.org/10.1103/PhysRev.85.180>.
- Max Born. Quantenmechanik der stoßvorgänge. *Zeitschrift fr Physik*, 38(11–12):803–827, November 1926. ISSN 1434-601X. doi:10.1007/bf01397184. URL <http://dx.doi.org/10.1007/BF01397184>.
- Sean Carroll. *Something deeply hidden*. Dutton, London, England, September 2019.
- Sean M. Carroll and Ashmeet Singh. *Mad-Dog Everettianism: Quantum Mechanics at Its Most Minimal*, page 95–104. Springer International Publishing, 2019. ISBN 9783030113018. doi:10.1007/978-3-030-11301-8_10. URL http://dx.doi.org/10.1007/978-3-030-11301-8_10.
- David Deutsch. Quantum theory of probability and decisions. *Proceedings of the Royal Society of London. Series A: Mathematical, Physical and Engineering Sciences*, 455(1988):3129–3137, August 1999. ISSN 1471-2946. doi:10.1098/rspa.1999.0443. URL <http://dx.doi.org/10.1098/rspa.1999.0443>.
- Hugh Everett. “relative state” formulation of quantum mechanics. *Reviews of Modern Physics*, 29(3):454–462, July 1957. ISSN 0034-6861. doi:10.1103/revmodphys.29.454. URL <http://dx.doi.org/10.1103/RevModPhys.29.454>.
- Kazuo Fujikawa. Path-integral measure for gauge-invariant fermion theories. *Physical Review Letters*, 42(18):1195–1198, April 1979. ISSN 0031-9007. doi:10.1103/physrevlett.42.1195. URL <http://dx.doi.org/10.1103/PhysRevLett.42.1195>.
- Andrew M. Gleason. Measures on the closed subspaces of a hilbert space. *Journal of Mathematics and Mechanics*, 6:885–893, 1957.
- Andrew M. Gleason. *Measures on the Closed Subspaces of a Hilbert Space*, page 123–133. Springer Netherlands, 1975. ISBN 9789401017954. doi:10.1007/978-94-010-1795-4_7. URL http://dx.doi.org/10.1007/978-94-010-1795-4_7.
- James Glimm and Arthur Jaffe. *Quantum physics*. Springer, New York, NY, 2 edition, May 1987.
- Sheldon Goldstein. *Typicality and Notions of Probability in Physics*, page 59–71. Springer Berlin Heidelberg, August 2011. ISBN 9783642213298. doi:10.1007/978-3-642-21329-8_4. URL http://dx.doi.org/10.1007/978-3-642-21329-8_4.
- Lucien Hardy. Quantum theory from five reasonable axioms, 2001.
- T. W. B. Kibble. Geometrization of quantum mechanics. *Communications in Mathematical Physics*, 65(2):189–201, 1979. doi:10.1007/BF01178148.
- A N Kolmogorov. *Foundations of the theory of probability: Second English edition*. Courier Dover Publications, March 2018.
- Hui-Hsiung Kuo. *Gaussian Measures in Banach Spaces*. Springer Berlin Heidelberg, 1975. ISBN 9783540375081. doi:10.1007/bfb0082007. URL <http://dx.doi.org/10.1007/BFb0082007>.
- John M. Lee. *Introduction to Smooth Manifolds*. Springer, New York, 2nd edition, 2012. ISBN 9781441999818.
- Yu. V. Prokhorov. Convergence of random processes and limit theorems in probability theory. *Theory of Probability & Its Applications*, 1(2):157–214, January 1956. ISSN 1095-7219. doi:10.1137/1101016. URL <http://dx.doi.org/10.1137/1101016>.
- Walter Rudin. *Real and Complex Analysis*. McGraw-Hill, New York, 3rd edition, 1987. ISBN 9780070542341.
- Charles T. Sebens and Sean M. Carroll. Self-locating uncertainty and the origin of probability in everettian quantum mechanics. *The British Journal for the Philosophy of Science*, 69(1):25–74, March 2018. ISSN 1464-3537. doi:10.1093/bjps/axw004. URL <http://dx.doi.org/10.1093/bjps/axw004>.
- Lev Vaidman. Why the many-worlds interpretation? *Quantum Reports*, 4(3):264–271, 2022. ISSN 2624-960X. doi:10.3390/quantum4030018. URL <https://www.mdpi.com/2624-960X/4/3/18>.
- John Von Neumann. *Mathematical foundations of quantum mechanics*. Princeton Landmarks in Mathematics and Physics. Princeton University Press, Princeton, NJ, October 1996.
- David Wallace. *The Emergent Multiverse: Quantum Theory according to the Everett Interpretation*. OUP Oxford, May 2012.

- Kenneth G. Wilson. Confinement of quarks. *Physical Review D*, 10(8):2445–2459, October 1974. ISSN 0556-2821. doi:10.1103/physrevd.10.2445. URL <http://dx.doi.org/10.1103/PhysRevD.10.2445>.
- Wojciech Hubert Zurek. Environment-assisted invariance, entanglement, and probabilities in quantum physics. *Physical Review Letters*, 90(12), March 2003. ISSN 1079-7114. doi:10.1103/physrevlett.90.120404. URL <http://dx.doi.org/10.1103/PhysRevLett.90.120404>.
- Gerard 't Hooft. *The Cellular Automaton Interpretation of Quantum Mechanics*. Springer International Publishing, 2016. ISBN 9783319412856. doi:10.1007/978-3-319-41285-6. URL <http://dx.doi.org/10.1007/978-3-319-41285-6>.